Math Inquirers

Extension Workbook

with answers

Level 5

Self-study | Ages 10-12



My name _____

About this book

Math Inquirers Extension Workbook, Level 5
Self-study practice with answers.

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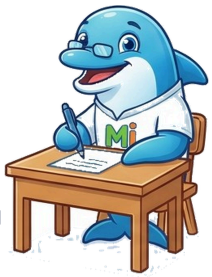
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- **Start here.** The main path. Do these first.
- **Think harder.** Longer problems. Hints are printed upside down.
- **Go further.** A weekend page. Big challenges, just for fun.
- **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Million puzzles**	6-7	8-9	10 and 12	11	☐
2	**Fraction feats**	13-14	15-16	17 and 19	18	☐
3	**Before and after**	20-21	22-23	24 and 26	25	☐
4	**Triangle areas**	27-28	29-30	31 and 33	32	☐
5	**Volume vault**	34-35	36-37	38 and 40	39	☐
6	**Rate racers**	41-42	43-44	45 and 47	46	☐
7	**Percentage puzzles**	48-49	50-51	52 and 54	53	☐
8	**Angle chasers**	55-56	57-58	59 and 61	60	☐
9	**Pattern generalisers**	62-63	64-65	66 and 68	67	☐
10	**Challenge day**	69-70	71-72	73 and 75	74	☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Numbers to 10 million

My goal. I can read, compare and round numbers up to 10 million. I can test both sides of a target and check every clue.

Read a big number in groups of three from the right. To find the number closest to a target, test the best number just below it and just above it.

Worked example

Use the cards to make a 7-digit number as close as you can to 3 000 000. Use every card once. A number cannot start with 0.

0	1	2	3	5	8	9

1. **Just below the target: start with 2. Then put the cards in order from greatest to smallest: 2 985 310.**
2. **Just above the target: start with 3. Then put the rest in order from smallest to greatest: 3 012 589.**
3. **Find each gap. 3 000 000 − 2 985 310 = 14 690. And 3 012 589 − 3 000 000 = 12 589.**
4. **Compare the gaps. 12 589 is smaller than 14 690, so the number above is closer.**

Answer: The closest number is 3 012 589.

Closest number

1. Write the best number just below the target.
2. Write the best number just above the target.
3. Find the gap for each. The smaller gap wins.

Place-value chart

M	HTh	TTh	Th	H	T	O
4	0	5	3	2	0	1

- The number is 4 053 201.
- M is millions. TTh is ten-thousands.

Your turn

Answer at the back of the book.

Try **Use the cards to make a 7-digit number as close as you can to 8 000 000. Use every card once. Write your number.** _____

0	1	3	4	6	7	9

Start here

Do these first. Take your time.

1 **Fill in the blanks.**

(a) Write the number: 6 million, 205 thousand and 30.

_____

(b) In 3 470 218, the digit 7 stands for _____

(c) How many thousands make 8 604 000? ☐ thousands

2 **Arrange the numbers in order. Begin with the smallest.**
4 096 213, 940 621, 4 906 213, 4 096 231

_____

3 **Round 7 849 520 to the nearest:**

(a) thousand. _____

(b) hundred thousand. _____

(c) million. _____

4 **Look at the cards. Use all 7 cards once in each answer. A number cannot start with 0.**

0	1	4	5	7	8	9

(a) The greatest multiple of 5 is _____

(b) The smallest number is _____

5 **Ben has the number below. He adds 2 hundred-thousands. Then he takes away 5 ten-thousands. What is Ben's new number?**

M	HTh	TTh	Th	H	T	O
6	4	8	2	3	1	5

_____

Start here

Keep going. Check your work.

6 **Estimate each answer by rounding. Circle one choice for each.**

(a) 3 985 × 52 = 20 722 could be right must be wrong

(b) 4 012 018 + 2 899 987 = 6 912 005 could be right must be wrong

(c) 8 120 × 295 = 2 395 400 could be right must be wrong

(d) 6 480 ÷ 16 = 4 050 could be right must be wrong

7 **Put ONE pair of brackets in each line to make it true. Write the new line.**

(a) 9 - 2 × 3 + 1 = 22 _____

(b) 9 - 2 × 3 + 1 = 1 _____

(c) 2 400 000 ÷ 300 + 100 × 2 = 12 000

_____

8 **A factory makes 1 250 000 bolts. It packs 500 bolts in each box. Each carton holds 24 boxes. How many full cartons can it fill?** ☐ **cartons**

9 **A number rounded to the nearest hundred is 8 700. Fill in the blanks.**



(a) The smallest such number is ☐

(b) The greatest such number is ☐

10 **Circle True or False.**

(a) 5 400 000 is 54 hundred thousands. True False

(b) 1 000 000 - 1 = 99 999. True False

(c) The greatest 6-digit number is 1 less than the smallest 7-digit number.
True False

Start here

Nearly there. You can do it.

11 Look at the table.

Town	People
Alder	1 204 500
Birch	1 240 500
Cedar	1 024 500
Dune	1 420 050

(a) Which town has the most people? ☐

(b) How many more people live in the town with the most people than in the town with the fewest? _____

12 Solve the riddle in the box. Check every clue when you finish.

Clue 1 I am a 6-digit number. All my digits are different.

Clue 2 I round to 470 000 when rounded to the nearest ten-thousand.

Clue 3 My thousands digit is 2.

Clue 4 My hundreds digit is twice my tens digit.

Clue 5 My ones digit is 0.

I am _____

13 A stadium has 48 000 seats. On Monday 3 100 people came. On Tuesday twice as many came. On Wednesday 1 250 fewer people came than on Tuesday. How many people came in all? _____ people

14 Circle every number that rounds to 5 000 000 when rounded to the nearest million.

4 499 999 4 500 000 5 499 999 5 500 000

Think harder

Hints are upside down. Try first!

1 **Solve the riddle in the box. Test the clues one at a time.**

Clue 1	**I am a 7-digit number. All my digits are different.**
Clue 2	**I round to 6 400 000 when rounded to the nearest hundred-thousand.**
Clue 3	**My ten-thousands digit is 1 less than my hundred-thousands digit.**
Clue 4	**My hundreds digit is 2 more than my thousands digit.**
Clue 5	**My tens digit is 0.**
Clue 6	**My digits add up to 33.**

I am _____

Hint. Clues 2 and 3 fix the first 3 digits. Then use Clue 6.

2 **Every number from 195 000 to 204 999 rounds to 200 000 when rounded to the nearest ten-thousand.**

(a) How many of these numbers have a digit sum of 4? ☐

Working: _____

Hint. Numbers from 195 000 to 199 999 have a digit sum over 4.

3 **Use each card once to make a 3-digit number and a 2-digit number. Multiply the 2 numbers. Make the product as great as you can.**



The product is _____

Working: _____

Hint. The greatest digits go first. Try both ways.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 5: Counting choices**
The first digit is never 0. How many 7-digit numbers have 7 different digits?

_____

Working: _____

Hint. Count the choices for the first digit, then the second.

2 **Ali writes down every whole number from 1 to 999. He adds up all the digits he wrote. What is the total?** _____

Working: _____

Hint. Write 7 as 007. Each digit 0 to 9 appears equally often.

3 **Use each of 3, 4, 7 and 9 once. You may use + − × ÷ and brackets. What is the number closest to 100 that you can make?** ☐

Working: _____

Hint. 9 × 11 is 99, but 11 is hard to make. Try 3 × 4 × 9.

4 **A printing press prints 2 400 sheets in 1 hour. It runs for 8 hours each day. A shop needs 1 000 000 sheets. How many days does the press need?** ☐ **days**

Working: _____

Hint. Sheets in one day first. Then divide. Is the answer a whole number?

✔ Check yourself

No hints. Try each one on your own.

1 **Look at the number 5 280 436.**

(a) The digit 2 stands for _____

(b) Rounded to the nearest 100 000, it is _____

Working: _____

2 **Build the 7-digit number nearest to 4 000 000. Use each card once.**

0	2	3	5	6	7	9

My number is _____

3 **Put ONE pair of brackets in 8 + 12 ÷ 4 × 2 = 10 to make it true. Write the new line.**

_____

Working: _____

4 **18 buses each carry 1 250 passengers in a week. How many passengers is that in 12 weeks?** _____

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Fractions: four operations

My goal. I can link fractions and division. I can find a fraction of a remainder, and work backwards with a bar model.

A fraction of a remainder is taken from what is left, not from the whole. First find what is left. Then take the fraction of that. A bar model with equal parts shows every step.

Worked example

A ribbon is 180 cm long. Mei uses 1/3 of it for a gift. Then she uses 1/4 of the ribbon that is left for a bow. How long is the ribbon that is left now?



1. **Cut the bar into 6 equal parts. Each part is 180 ÷ 6 = 30 cm. The gift uses 1/3, which is 2 parts.**
2. **The part left is 4 parts. The bow uses 1/4 of those 4 parts, which is 1 part.**
3. **The ribbon that is left is 6 - 2 - 1 = 3 parts.**
4. **3 parts × 30 cm = 90 cm.**

Answer: The ribbon that is left is 90 cm long.

How to

1. Draw a bar. Cut it into equal small parts.
2. Take the first fraction. Find what is left.
3. Take the next fraction of what is left, not of the whole.

Remember

- a ÷ b = a/b. For example, 3 ÷ 4 = 3/4.
- A unit fraction has 1 on top, such as 1/5.
- To add or subtract, use the same bottom number.

Your turn

Answer at the back of the book.

Try **Mei has 120 beads. She uses 1/4 of them for a bracelet. Then she uses 2/3 of the rest for a necklace. How many beads are left?** ☐

Start here

Do these first. Take your time.

1 **Fill in the blanks. Write each answer as a fraction.**

(a) 7 pizzas are shared equally among 10 children. Each child gets ☐ of a pizza.

(b) A 3 m rope is cut into 8 equal pieces. Each piece is ☐ m long.

(c) 9 ÷ 4 = ☐ (write a mixed number)

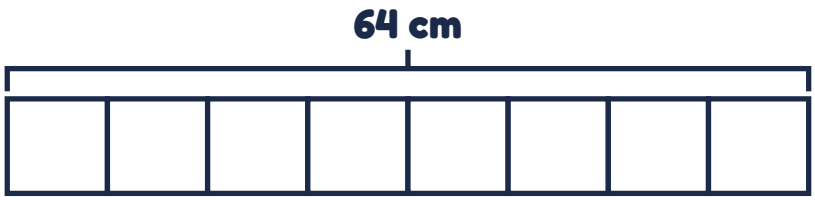
2 **Circle the greater amount in each pair.**

(a) 4/5 or 5/6 4/5 5/6

(b) 5/9 or 4/7 5/9 4/7

(c) 3/4 of 20 or 2/3 of 24 3/4 of 20 2/3 of 24

3 **A bar of 8 equal parts is 64 cm long. Find the fractions of the bar.**



(a) 3/8 of 64 cm is ☐ cm.

(b) 5/8 of 64 cm is ☐ cm.

(c) 1/4 of 64 cm is ☐ cm.

4 **Find each answer. Write it in its simplest form.**

(a) 2/3 + 1/4 = ☐

(b) 5/6 - 3/8 = ☐

(c) 3/4 - 1/6 = ☐

Start here

Keep going. Check your work.

5 Ravi walks 3/4 km each day. He does this for 6 days. How far does he walk in all?

Write a mixed number. ☐ km

6 Sam has 72 cards. He gives 1/3 of them to Mei. Then he gives 3/4 of the cards that are left to Ben. How many cards does Sam have now? ☐

7 Write 7/12 as a sum of 2 different unit fractions. A unit fraction has 1 on top.

Take away 1/3 first: 7/12 − 1/3 = ☐. So 7/12 = 1/3 + ☐.

8 Look at the pattern: 1/3, 2/5, 3/7, 4/9, ...

(a) The next 2 fractions are ☐ and ☐.

(b) The 10th fraction is ☐.

(c) 13/27 is the ☐ th fraction.

9 Ali reads 2/5 of a book on Saturday and 1/4 of the book on Sunday. What fraction of the book is not read yet? ☐

10 Mark Mei's points A and B on the number line. Write each as a mixed number.



(a) A = ☐

(b) B = ☐

Start here

Nearly there. You can do it.

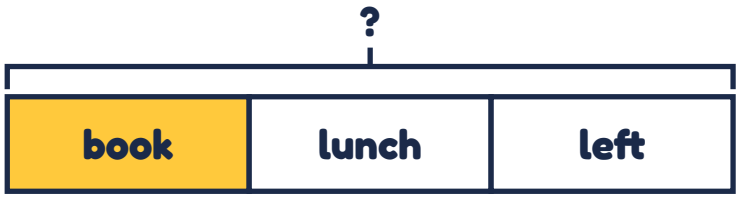
11 **Multiply. Write each answer in its simplest form.**

(a) 2/3 × 3/8 = ☐

(b) 3/5 of 2/3 = ☐

(c) 4/9 × 3 = ☐ (mixed number)

12 **Mei buys a book with 1/3 of her money. Then she pays 1/2 of the rest for lunch. She has $18 left. Use the model.**



(a) Lunch cost ☐ dollars.

(b) Mei had ☐ dollars at first.

Working: _____

13 **In a class of 40 children, 3/8 walk to school and 1/4 take the bus. The rest cycle. How many children cycle?** ☐

Working: _____

14 **Circle True or False.**

(a) 3 ÷ 5 = 5/3 True False

(b) 7/8 is greater than 5/6. True False

(c) 1/4 of 1/2 is 1/8. True False

Think harder

Hints are upside down. Try first!

1 **Write 2/5 as the sum of 2 different unit fractions. Try 1/3 first, then 1/4, then 1/5. Stop when the first fraction is less than half of 2/5.**

2/5 = ☐ + ☐

Working: _____

_____

_____

Hint. Take away a unit fraction. Is what is left a unit fraction?

2 **A shop sells 1/3 of its eggs in the morning. It sells 1/4 of the rest in the afternoon. In the evening it sells 2/5 of the eggs then left. At the end 36 eggs are left. How many eggs did the shop have at first?** ☐

Working: _____

_____

_____

Hint. Find the fraction of ALL the eggs that is left at each step.

3 **Look at the cards. Make 2 fractions and add them. Each card is used once. The fractions need not be proper.**

2	3	5	7

(a) The greatest sum is ☐

(b) The smallest sum is ☐

Working: _____

_____

_____

Hint. A big top and a small bottom make a big fraction.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 5: Dividing a fraction**

5/6 of a cake is shared equally among 10 children. What fraction of the whole cake does each child get? ☐

Working: _____

_____

Hint. Cut each sixth into 2 equal parts. How many parts are there?

2 **Write 1/6 as the sum of 2 different unit fractions. How many ways are there?**

☐ **ways**

Working: _____

_____

Hint. Test 1/7, 1/8, 1/9 … Stop when the first is under half of 1/6.

3 **A ball falls from a height. Each time it bounces up to 3/4 of the height it fell from. After the 3rd bounce it reaches 27 cm.**

(a) The ball fell from ☐ cm at first.

(b) The first bounce that is under half the first height is bounce ☐.

Working: _____

Hint. After 3 bounces the height is 3/4 of 3/4 of 3/4 of the start.

4 **Find 1/3 + 1/15 + 1/35 + 1/63.** ☐

Working: _____

_____

Hint. 1/3 = 1/2 × (1 - 1/3). Write each as a difference.

✔ Check yourself

No hints. Try each one on your own.

1 **5 pizzas are shared equally among 6 children. Write the answers as fractions.**

(a) Each child gets ☐ of a pizza. (b) 7 ÷ 3 = ☐ (mixed number)

Working: _____

2 **Find each answer in its simplest form.**

(a) 7/8 − 1/3 = ☐ (b) 2/5 + 3/10 = ☐

Working: _____

3 **Kumar has 120 marbles. He gives 1/5 of them to Ali. He gives 3/8 of the rest to Wei. How many marbles does Kumar have now?** ☐

Working: _____

4 **Siti uses 3/4 of a bag of flour for bread and 1/6 of the bag for a cake. What fraction of the bag is left?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

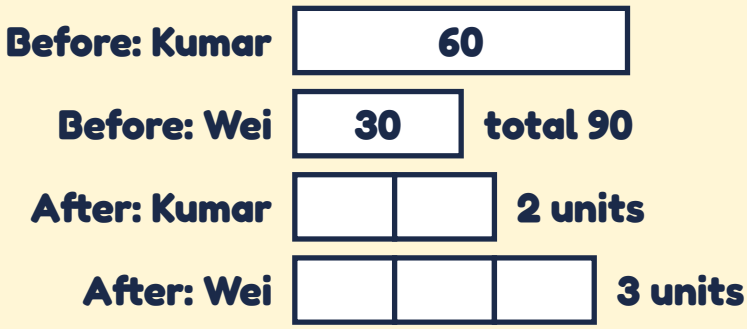
Learn it: Fraction word problems

My goal. I can use bar models for before and after. I ask what stays the same: the total, the difference or one amount.

Draw a model for the Before and a model for the After. Ask first: what stays the same?

Worked example

Kumar has 60 cards. Wei has 30 cards. Kumar gives some cards to Wei. Then Kumar has 2/3 as many cards as Wei. How many cards does Kumar give?



1. Kumar gives cards to Wei, so the total stays the same: 60 + 30 = 90.
2. After: Kumar has 2 units and Wei has 3 units. That is 5 units in all.
3. 5 units = 90, so 1 unit = 18. Kumar now has 2 × 18 = 36.
4. Kumar gives 60 − 36 = 24 cards.

Answer: Kumar gives 24 cards.

How to

1. Draw the Before bars. Draw the After bars.
2. Ask what stays the same. Make that part match.
3. Find 1 unit. Then answer the question.

What stays the same?

- A gives to B: the total.
- Both get or lose the same: the difference.
- Only one thing changes: the other amount.

Your turn

Answer at the back of the book.

Try **Jason has 70 stickers. Amelia has 50 stickers. Jason gives some stickers to Amelia. Then Jason has 3/5 as many stickers as Amelia. How many stickers does Jason give?** ☐

Start here

Do these first. Take your time.

1 **Aisha has $80. A book takes 1/4 of her money. A pen takes 1/3 of the money that is then left.**

(a) The book costs $ ☐

(b) The pen costs $ ☐

(c) Aisha has $ ☐ left.

2 **Lily has 22 counters. Ravi has 14 counters. Lily gives 5 counters to Ravi.**

(a) Lily has ☐ counters now.

(b) Ravi has ☐ counters now.

(c) The total is ☐ before and after.

3 **Ben is 12 years old. His aunt is 41 years old.**

(a) In 6 years Ben will be ☐ and his aunt will be ☐.

(b) Now the aunt is ☐ years older than Ben. In 6 years she is ☐ years older.

4 **Ali has 38 stamps. Mei has 24 stamps. Ali gives some stamps to Mei. Then each child has an equal number of stamps. How many stamps does Ali give?** ☐

5 **There are 48 children. 5/12 of them wear glasses. How many more children do not wear glasses than wear glasses?** ☐

Start here

Keep going. Check your work.

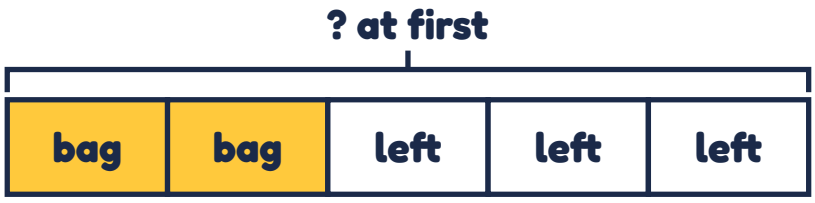
6 **A bag has 15 green beads and 9 yellow beads. Then 6 more yellow beads are put in.**

(a) The beads that did not change are green yellow

(b) At first, green beads are ☐ of the beads. (simplest form)

(c) Now, green beads are ☐ of the beads. (simplest form)

7 **Ray spent 2/5 of his money on a bag. He has $45 left. Use the bar model.**



(a) 1 part is $ ☐

(b) Ray had $ ☐ at first.

8 **A pen costs 1/3 as much as a book. The pen and the book cost $28 together.**

(a) The book costs $ ☐

(b) The pen costs $ ☐

9 **Mei has 3/4 as many stickers as Siti. They have 56 stickers altogether.**

(a) Siti has ☐ stickers.

(b) Mei has ☐ stickers.

Start here

Nearly there. You can do it.

10 **What stays the same? Circle one choice for each.**

(a) Ali gives some marbles to Ben. the total the difference

(b) Ravi and Ali are each given $5. the total the difference

(c) Red beads are added to a jar of red and blue beads.
the red beads the blue beads

11 **Ravi has $45. Ben has $21. Their mother gives each boy $15.**

(a) Ravi has $ ☐ now.

(b) Ben has $ ☐ now.

(c) Ravi has $ ☐ more than Ben now.

12 **Siti buys a shirt with 1/4 of her money. Shoes take 1/3 of the rest. She keeps $30.**
How much money did Siti start with? $ ☐

Working: _____

13 **A jug was 5/6 full of juice. After 250 ml was poured out, it was 1/2 full. How much juice does the jug hold when it is full?** ☐ **ml**

Working: _____

14 **Circle True or False.**

(a) When Ali gives Ben some stickers, the total number of stickers changes.
True False

(b) When 2 children are each given $5, the difference between their money stays the same. True False

(c) If Mei has 1/2 as many cards as Ali, then Ali has 2 times as many cards as Mei. True False

Think harder

Hints are upside down. Try first!

1 **On Monday Siti uses 2/5 of her money. On Tuesday she uses 1/3 of what is left. In all she uses $90. How much money did Siti start with? $ ☐**

Working: _____

Hint. Cut the whole into 15 equal units. Count the units spent.

2 **Siti and Mei have 120 beads altogether. Siti gives 10 beads to Mei. Then Siti has 3/5 as many beads as Mei. How many beads did Siti have at first? ☐**

Working: _____

Hint. The total stays the same. Find each child's beads after.

3 **Ravi is 38 years old. His son is 10 years old. In how many years will the son be 1/3 of his father's age? ☐ years**

Working: _____

Hint. The difference in their ages never changes. How many units is it?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Ali, Ben and Cara have 150 stickers altogether. Ali has twice as many stickers as Ben. Ben has 10 more than Cara.**

(a) Cara has ☐ stickers.

(b) Ben has ☐ stickers.

(c) Ali has ☐ stickers.

Working: _____

Hint. Call Cara's stickers 1 unit. Draw bars for Ben and for Ali.

2 **In a hall, 2/5 of the people are children. Then 15 adults leave and 15 children come in. Now 1/2 of the people are children. How many people are in the hall?**

☐

Working: _____

Hint. The total stays the same. How much more of the total is children?

3 **A jar holds red beads and white beads. Red beads make up 2/7 of the jar. Then 24 red beads are taken out. Now red beads make up 1/6 of the jar. How many beads were in the jar at first?** ☐

Working: _____

Hint. The white beads do not change. What fraction of each total is white?

4 **Ali and Ben have 120 marbles altogether. Ali gives 1/3 of his marbles to Ben. Then Ben has 3 times as many marbles as Ali. How many marbles did Ali have at first?** ☐

Working: _____

Hint. After giving, Ali has 2/3 of his marbles. Ben has 3 times that.

✔ Check yourself

No hints. Try each one on your own.

1 **Jenny pays 1/5 of her money for a book. A bag takes 1/2 of the rest. She keeps $24. How much money did Jenny start with? $ ☐**

Working: _____

2 **Jamie has 46 cards. Jason has 30 cards. Jamie gives some cards to Jason. Then they have equal numbers. How many cards does Jamie give? ☐**

Working: _____

3 **Amelia has $50. Ray has $30. They are each given the same amount. Then Ray has 3/4 as much as Amelia. How much is each given? $ ☐**

Working: _____

4 **A jar has 18 blue beads and 12 yellow beads. Then 20 yellow beads are put in.**

(a) The beads that did not change are blue yellow

(b) The fraction of beads that are blue is now ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

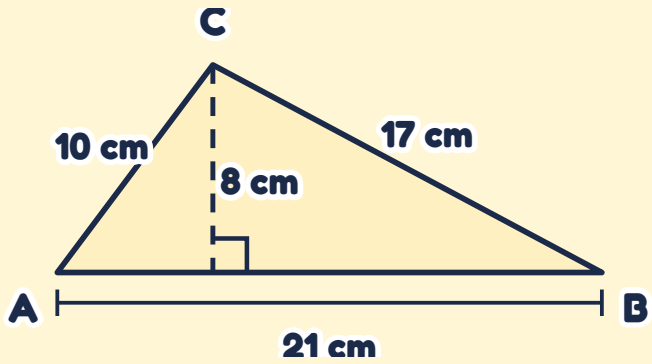
Learn it: Area of a triangle

My goal. I can find the area of a triangle with the height that meets the base at a right angle. I can find areas of shapes made from triangles.

The area of a triangle is 1/2 × base × height. The height meets the base at a right angle. A slanting side is not the height, unless the triangle has a right angle there.

Worked example

Find the area of triangle ABC. The dashed line is the height.



1. Choose the base: AB = 21 cm. The height meets AB at a right angle: it is 8 cm.
2. The sides AC = 10 cm and BC = 17 cm are slanting. They are not heights, so we do not use them.
3. Area = 1/2 × base × height = 1/2 × 21 × 8.
4. 21 × 8 = 168 and half of 168 is 84.

Answer: The area of triangle ABC is 84 cm².

How to

1. Find a base and the height that meets it at a right angle.
2. Ignore the slanting sides.
3. Multiply base × height. Then halve it.

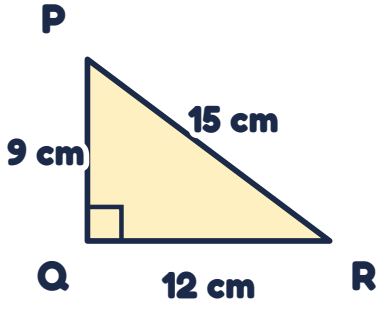
Your turn

Answer at the back of the book.

Try

Triangle PQR has a right angle at Q. PQ = 9 cm, QR = 12 cm and PR = 15 cm. Find its area.

_____ cm²



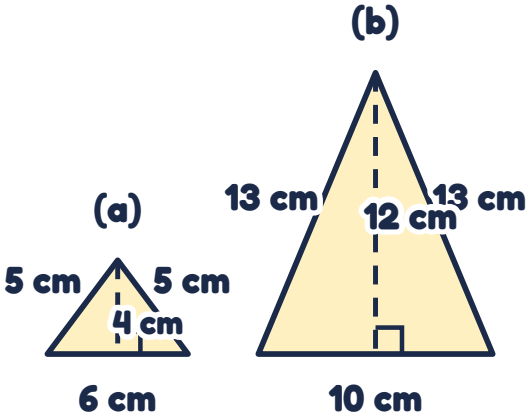
Start here

Do these first. Take your time.

1 **Find the area of each triangle. The dashed line is the height.**

(a) Triangle (a): ☐ cm²

(b) Triangle (b): ☐ cm²



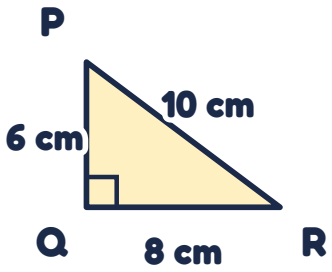
2 **Triangle PQR has a right angle at Q.**

(a) The area is ☐ cm².

(b) With QR as the base, the height is ☐ cm.

(c) With PQ as the base, the height is the side

QR PR



3 **The area of a triangle is 1/2 × base × height. Find the missing length.**

(a) Area 36 cm² and base 9 cm. The height is ☐ cm.

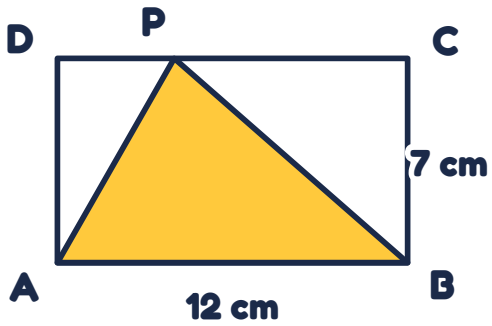
(b) Area 45 cm² and height 9 cm. The base is ☐ cm.

4 **ABCD is a rectangle. Point P lies on the top edge DC.**

(a) The shaded triangle ABP has area ☐ cm².

(b) P slides along DC. The area

gets greater gets smaller is the same



Start here

Keep going. Check your work.

5

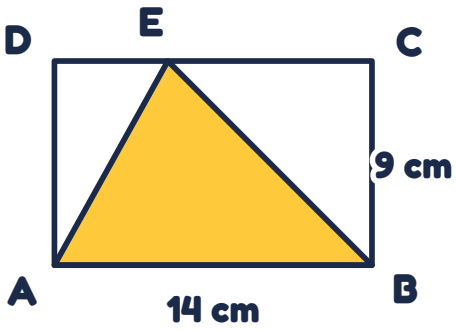
A square has sides of 8 cm. A triangle stands on its top edge, with the same 8 cm as its base. The triangle is 5 cm tall. What is the total area? ☐ cm²



6 ABCD is a rectangle. Point E lies on the top edge DC. Triangle ABE is shaded.

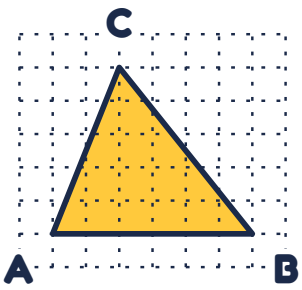
(a) The shaded area is ☐ cm².

(b) The area that is not shaded is ☐ cm².



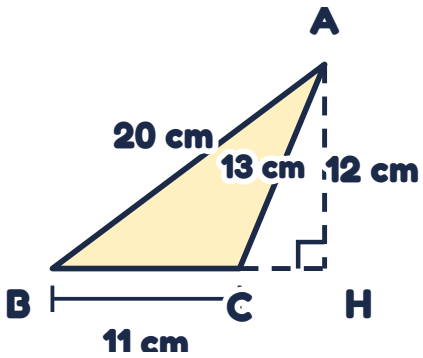
7

Each square on the grid is 1 cm². Find the area of the triangle. ☐ cm²



8

Look at triangle ABC. Its height AH is outside the triangle. Find the area of triangle ABC. ☐ cm²



Start here

Nearly there. You can do it.

9 **The areas of a square and a triangle are equal. The square has sides of 10 cm. The base of the triangle is 25 cm.**

(a) The area of the square is ☐ cm².

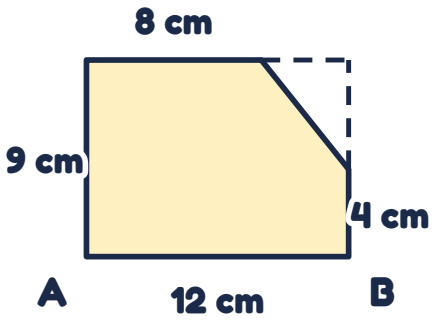
(b) The height of the triangle is ☐ cm.

Working: _____

10

The shape is a rectangle with a corner cut off.

Find its area. ☐ cm²



Working: _____

11 **A triangle has an area of 9 cm². Its base and height are whole numbers of cm.**

(a) How many different pairs of base and height are possible? ☐

(b) The pair with a base of 3 cm has a height of ☐ cm.

Working: _____

12 **Circle True or False.**

(a) The height of a triangle meets its base at a right angle. True False

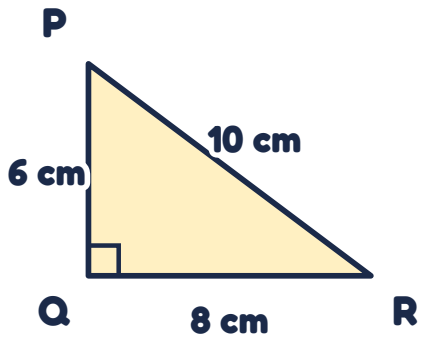
(b) Any slanting side of a triangle can be used as the height. True False

(c) A triangle with base 10 cm and height 6 cm has the same area as a triangle with base 6 cm and height 10 cm. True False

Think harder

Hints are upside down. Try first!

1 Triangle PQR has a right angle at Q. Its sides are 6 cm, 8 cm and 10 cm. Now use the 10 cm side PR as the base. The height from Q to PR is ☐ cm.

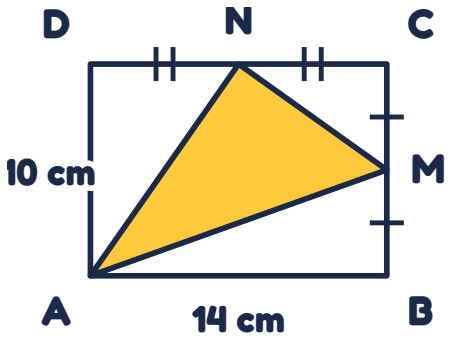


Working: _____

Hint. Find the area first. Then use the 10 cm side as the base.

2 ABCD is a rectangle, 14 cm by 10 cm. M is the middle of BC. N is the middle of CD. How much area does the shaded triangle AMN cover?

☐ cm²



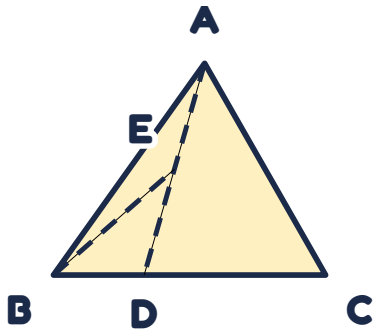
Working: _____

Hint. Take the 3 white triangles away from the rectangle.

3 The area of triangle ABC is 72 cm². D is on BC and BD is 1/3 of BC. E is the middle of AD.

(a) The area of triangle ABD is ☐ cm².

(b) The area of triangle BDE is ☐ cm².



Working: _____

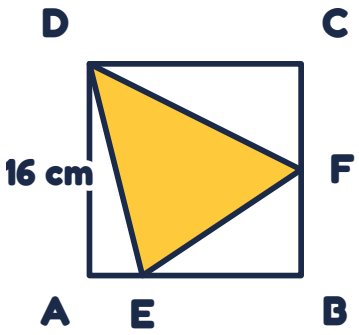
Hint. ABD and ABC have the same height. Compare their bases.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1

ABCD is a square with sides of 16 cm. E lies on AB, 4 cm from A. F is the middle of BC. What is the area of triangle DEF? ☐ **cm²**



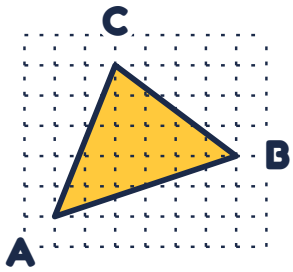
Working: _____

Hint. Take away the 3 white right-angled triangles.

2

Each square on the grid is 1 cm². The corners of the triangle are on grid points. Find its area.

☐ **cm²**



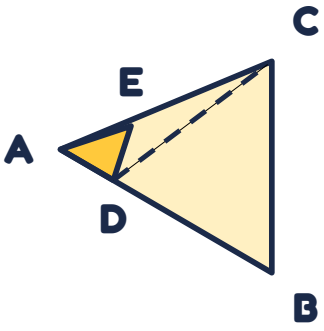
Working: _____

Hint. Draw a rectangle round the triangle. Take away 3 triangles.

3

Beyond Level 5: Corner triangles

Triangle ABC has an area of 96 cm². D is on AB with AD = 1/4 of AB. E is on AC with AE = 1/3 of AC. Find the area of triangle ADE. ☐ **cm²**



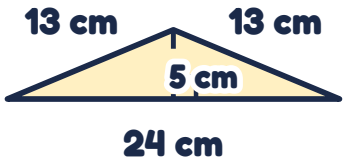
Working: _____

Hint. Join D to C. Compare ADC with ABC first.

Check yourself

No hints. Try each one on your own.

1 Find the area of the triangle. ☐ cm²

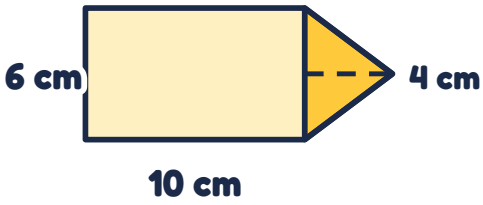


2 A triangle has an area of 54 cm². Its base is 12 cm. Find its height. ☐ cm

Working: _____

3 Find the area of the whole shape.

☐ cm²



4 A garden is a triangle with a base of 18 m and a height of 10 m. 1 bag of seed covers 6 m². How many bags are needed to cover the whole garden? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

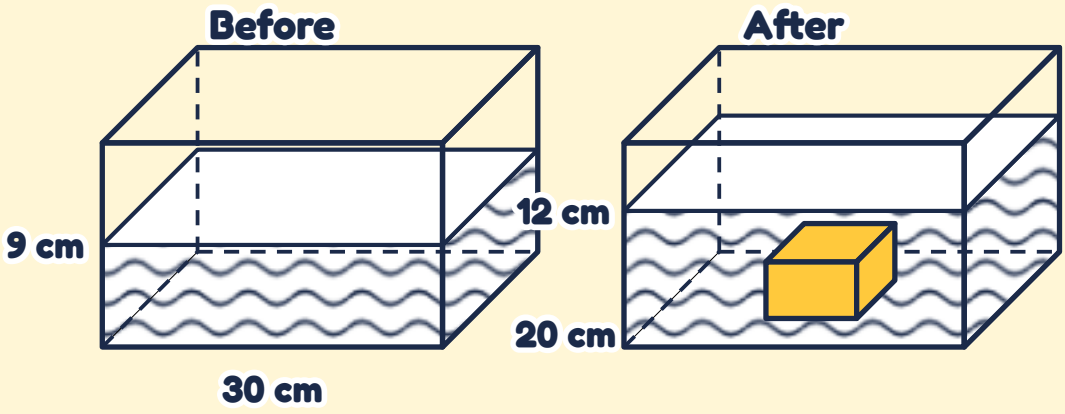
Learn it: Volume

My goal. I can find the volume of a cuboid and the volume of water in a tank. I can find the volume of a stone from how much the water rises.

Volume is the space inside a solid. For a cuboid, volume = length × breadth × height. 1 litre is 1000 cm³. A stone that sinks pushes up the water: the rise × the base area is the stone's volume.

Worked example

The floor of a tank is 30 cm × 20 cm. Water fills it to a depth of 9 cm. A stone sinks and the level rises to 12 cm. Find the volume of the stone.



1. The water rises 12 − 9 = 3 cm. The stone pushes the water up.
2. The stone's volume equals the extra water, a block 30 × 20 × 3 cm.
3. Volume = 30 × 20 × 3. First 30 × 20 = 600.
4. 600 × 3 = 1800.

Answer: The volume of the stone is 1800 cm³.

How to

1. Find the rise: new depth − old depth.
2. Volume = base length × base width × rise.
3. Write the unit: cm³. 1000 cm³ = 1 litre.

Your turn

Answer at the back of the book.

Try A tank floor is 25 cm × 20 cm. Water stands 10 cm deep. A stone of volume 1500 cm³ sinks in it. Find the new depth of the water. ☐ cm

Start here

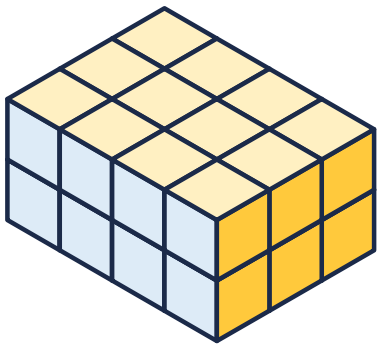
Do these first. Take your time.

1 **A box is built from 1 cm cubes. It has 4 cubes along the length, 3 along the width and 2 layers.**

(a) 1 layer has ☐ cubes.

(b) There are ☐ layers.

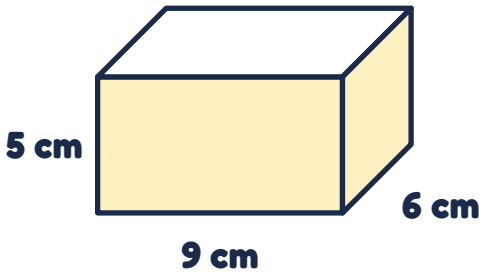
(c) The box has ☐ cubes. Its volume is ☐ cm^3.



2 **A cuboid measures 9 cm × 6 cm × 5 cm.**

(a) The volume of the cuboid is ☐ cm^3.

(b) A cube has edges of 7 cm. Its volume is ☐ cm^3.



3 **Find the missing length of each cuboid.**

(a) The base is 8 cm by 5 cm. The volume is 200 cm^3. The height is ☐ cm.

(b) The base is a square. The height is 4 cm. The volume is 144 cm^3. Each side of the base is ☐ cm.

4 **Fill in the blanks. Remember 1 litre = 1000 ml = 1000 cm^3.**

(a) 3.5 litres = ☐ cm^3

(b) 12 000 cm^3 = ☐ litres

(c) A tank is 50 cm long, 40 cm wide and 30 cm high. It holds ☐ litres when full.

Start here

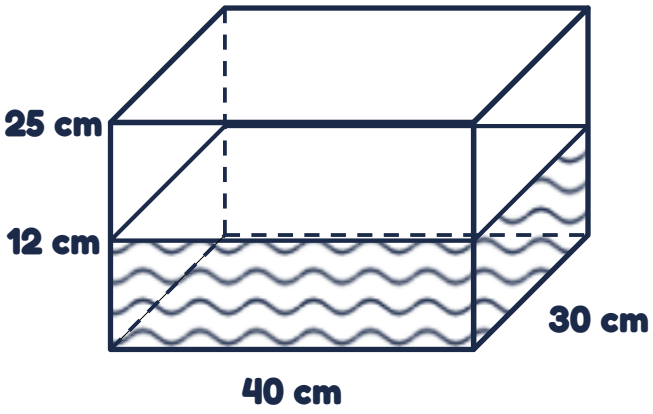
Keep going. Check your work.

5 **A tank measures 40 cm × 30 cm × 25 cm. It holds water 12 cm deep.**

(a) The water has a volume of ☐ cm^3.

(b) That is ☐ litres.

(c) To fill the tank to the top, ☐ litres more are needed.



6 **A tank floor is 50 cm × 20 cm. The water level is 6 cm. Then 4 litres are poured in.**

(a) The water rises ☐ cm.

(b) The water is now ☐ cm deep.

7 **The floor of a tank is 24 cm × 15 cm and the water is 8 cm deep. A stone sinks and the water level reaches 11 cm. Find the stone's volume.** ☐ cm^3

8 **A tank has a base 60 cm by 30 cm. The water is 20 cm deep. Then 9 litres of water are taken out.**

(a) The water level falls ☐ cm.

(b) The water is now ☐ cm deep.

9 **Cuboids are made with 30 cubes, using all the cubes each time. The edges are whole numbers of cm. A cuboid that is turned round is not new.**

(a) There are ☐ different cuboids.

(b) The cuboid with the smallest total of the 3 edges has a total of ☐ cm.

Start here

Nearly there. You can do it.

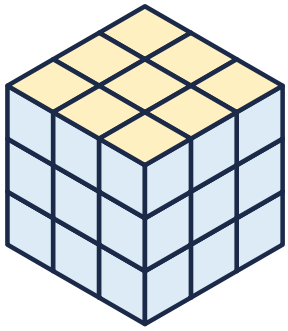
10 **A big cube is made from 27 small cubes (3 by 3 by 3). All its outside is painted. Then it is taken apart.**

(a) Cubes with 3 painted faces: ☐

(b) Cubes with 2 painted faces: ☐

(c) Cubes with 1 painted face: ☐

(d) Cubes with no paint: ☐



11 **Cubes with edges of 6 cm are packed in neat layers in a carton 32 cm long, 20 cm wide and 14 cm high. The cubes cannot be cut. How many cubes fit in the carton?** ☐

12 **A cube-shaped container with edges of 30 cm is full of water. All the water goes into an empty tank with a 60 cm × 45 cm floor. Find the depth of the water in the tank.** ☐ **cm**

13 **A tank is 2/5 full of water. Then 18 litres more go in and the tank is 7/10 full. Find the capacity of the full tank.** ☐ **litres**

Working: _____

14 **Circle True or False.**

(a) 1 litre of water has a volume of 1000 cm^3. True False

(b) A cuboid 2 cm × 3 cm × 4 cm has the same volume as a cube with edges of 3 cm. True False

(c) If all 3 edges of a cuboid are doubled, its volume is doubled.
True False

Think harder

Hints are upside down. Try first!

1 **A big cube is made of small cubes. Its outside is painted. Then it is taken apart. 54 small cubes have exactly 1 painted face. How many small cubes make the big cube?** ☐

Working: _____

_____

_____

Hint. Each face has the same number of 1-face cubes. How many?

2 **Tank A has a floor 40 cm × 30 cm. Its water is 18 cm deep. Tank B has a floor 30 cm × 20 cm and starts empty. Water is poured from A into B until the 2 tanks have equal depths. Find that depth.** ☐ **cm**

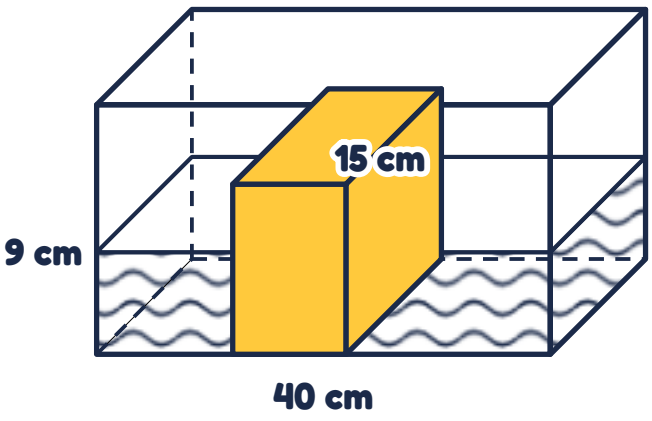
Working: _____

_____

_____

Hint. Add the 2 base areas. The same volume of water spreads over both.

3 **A tank floor is 40 cm × 20 cm and the water is 9 cm deep. A block 10 cm × 20 cm × 15 cm stands upright on the floor. Find the new water depth.** ☐ **cm**



Working: _____

Hint. Water cannot go where the block is. Find the free floor area.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 5: Cubes with tunnels**

A solid cube has edges of 6 cm. A tunnel 2 cm by 2 cm goes straight through the middle from the front to the back. 2 more tunnels of the same size go through the middle from left to right and from top to bottom. What is the volume of the solid that is left? ☐ **cm³**

Working: _____

Hint. Count the tunnel cubes. The middle cubes are in all 3 tunnels.

2 **A tank measures 60 cm × 50 cm × 40 cm and is empty. A tap pours in 9 litres each minute. A leak loses 1 litre each minute. After how many minutes is the tank full?**

☐

Working: _____

Hint. Find the capacity. Then the net litres gained each minute.

3 **The areas of 3 faces of a cuboid that meet at one corner are 24 cm², 36 cm² and 54 cm². Find the volume of the cuboid.** ☐ **cm³**

Working: _____

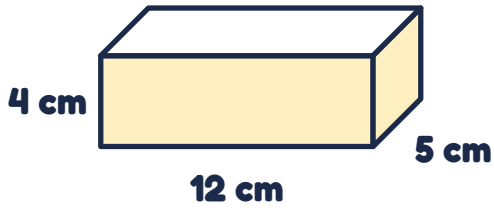
Hint. Guess 3 edges whose products in pairs are 24, 36 and 54.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the volume of the cuboid.**

☐ cm³



2 **A tank floor is 50 cm × 30 cm and the tank holds 18 litres. Find the water depth.**

☐ cm

Working: _____

3 **A cube is made of 64 small cubes (4 by 4 by 4). Its outside is painted. Count the small cubes with exactly 2 painted faces.** ☐

Working: _____

4 **A tank floor is 20 cm × 15 cm and the water is 10 cm deep. A stone sinks and the water rises to 14 cm. Find the volume of the stone.** ☐ cm³

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Rate

My goal. I can find a rate for 1 unit and use it to scale up. I can compare buys, and add rates when taps or machines work together.

A rate tells how much for 1 unit, such as litres in 1 minute. Find the amount for 1 unit first. Then multiply to scale up, or divide to find how long.

Worked example

A printer prints 90 pages in 6 minutes. How many pages does it print in 15 minutes? How many minutes does it need for 300 pages?

Minutes	Pages
6	90
1	15
15	225
20	300

1. **Find the rate: 90 ÷ 6 = 15 pages in 1 minute.**
2. **In 15 minutes: 15 × 15 = 225 pages.**
3. **For 300 pages: 300 ÷ 15 = 20 minutes.**
4. **Check: 20 minutes × 15 pages = 300 pages.**

Answer: It prints 225 pages in 15 minutes. It needs 20 minutes for 300 pages.

How to

1. Find the amount for 1 unit.
2. Multiply to scale up. Divide to find how long.
3. Check with the other way round.

Better buy

- Find the cost of 1 item.
- Or compare the cost of the same number of items.

Your turn

Answer at the back of the book.

Try **A machine fills 24 bottles in 4 minutes. How many bottles does it fill in 11 minutes?** ☐

Start here

Do these first. Take your time.

1 A tap fills 18 litres in 3 minutes. Fill in the table. Write the missing numbers.

Minutes	Litres
3	18
1	
5	
8	

2 6 pens cost $4.20.

(a) 1 pen costs $ ☐

(b) 9 pens cost $ ☐

3 An oven bakes 36 rolls in 4 minutes.

(a) It bakes ☐ rolls in 1 minute.

(b) In 25 minutes it bakes ☐ rolls.

(c) It needs ☐ minutes to bake 270 rolls.

4 Pack A has 5 pencils for $2.50. Pack B has 8 pencils for $3.60.

(a) Pack A costs $ ☐ for 1 pencil.

(b) Pack B costs $ ☐ for 1 pencil.

(c) The better buy is Pack A Pack B

Start here

Keep going. Check your work.

5 **Shop X sells 3 cans for $2.40. Shop Y sells 5 cans for $3.75. Mei needs 15 cans.**

(a) 15 cans at Shop X cost $ ☐

(b) 15 cans at Shop Y cost $ ☐

(c) The cheaper shop is Shop X Shop Y

6 **Tap A fills 14 litres a minute. Tap B fills 6 litres a minute. Both taps are on.**

(a) Together they fill ☐ litres in 1 minute.

(b) They need ☐ minutes to fill a 200 litre tank.

7 **A pump puts 25 litres a minute into a tank. A drain lets out 10 litres a minute. Both are on. In how many minutes is a 300 litre tank full?** ☐

Working: _____

8 **Printer P makes 40 pages each minute. Printer Q makes 20 pages each minute. Together they make one job of 900 pages. How many minutes does the job take?**

☐

Working: _____

9 **The first hour of parking costs $2.00. Each extra 30 minutes costs $0.80, and a part of 30 minutes counts as a whole one. Mei parks for 2 hours 20 minutes. How much does she pay? $** ☐

Working: _____

Start here

Nearly there. You can do it.

10 **A tank has 240 litres of water. A pump takes out 15 litres each minute.**

(a) After 9 minutes, ☐ litres are left.

(b) The tank is empty after ☐ minutes.

11 **A tap drips 4 ml of water every 10 seconds.**

(a) In 1 minute it drips ☐ ml.

(b) In 1 hour it drips ☐ ml.

(c) That is ☐ litres in 1 hour.

12 **Circle True or False.**

(a) If 6 cans cost $3.60, then 12 cans cost $7.20. True False

(b) A larger pack is always the better buy. True False

(c) Tap A fills 5 litres a minute. Tap B fills 7 litres a minute. Together they fill 12 litres a minute. True False

13 **Gym plan A costs $20 a month and then $4 for each visit. Plan B costs $8 for each visit and has no monthly fee. For how many visits in a month do the 2 plans cost the same?** ☐

Working: _____

14 **A pipe delivers 1.5 litres of water every 20 seconds. How many litres does it deliver in 5 minutes?** ☐ **litres**

Working: _____

Think harder

Hints are upside down. Try first!

1 **Pen packs: A has 5 pens for $3.00, B has 8 pens for $4.70 and C has 12 pens for $6.90. Mei needs exactly 32 pens and may mix packs. Find the least total cost. $ ☐**

Working: _____

Hint. List every way to make 32 pens with whole packs.

2 **Alone, printer P needs 12 minutes for a job. Alone, printer Q needs 20 minutes for the same job. Find the time for the job when both work. ☐ minutes**

Working: _____

Hint. Pretend the job has 60 pages. How many pages a minute for each?

3 **4 printers print 480 sheets in 6 minutes. All the printers work at the same rate. How many sheets do 6 printers print in 5 minutes? ☐**

Working: _____

Hint. Find the sheets for 1 printer in 1 minute first.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 5: Average rate**

A tank holds 120 litres. The first 60 litres go in at 6 litres a minute and the second 60 litres at 12 litres a minute. Average rate = total litres ÷ total minutes.

Find the average rate for the whole tank. ☐ litres a minute

Working: _____

_____

_____

_____

Hint. Find the minutes for each half first. Then add the minutes.

2 **Tank A has 90 litres. A pump takes 5 litres out of it each minute. Tank B has 10 litres. A tap puts 3 litres into it each minute. After how many minutes do the 2 tanks have the same amount of water? ☐ minutes**

Working: _____

_____

_____

_____

Hint. How many litres does the gap between the tanks close each minute?

3 **Tap A fills 8 litres a minute. Tap B fills 12 litres a minute. Tap A is turned on at the start. Tap B is turned on 5 minutes later. How many minutes after the start is a 200 litre tank full? ☐ minutes**

Working: _____

_____

_____

_____

Hint. How many litres has tap A put in before tap B starts?

✔ Check yourself

No hints. Try each one on your own.

1 **A copier makes 54 copies in 3 minutes. How many in 8 minutes?** ☐

Working: _____

2 **5 mangoes cost $4.00. 8 mangoes cost $6.00.**

(a) The better buy is 5 mangoes 8 mangoes

(b) It costs $ ☐ for 1 mango.

Working: _____

3 **Tap A fills 9 litres a minute, tap B fills 11 and a leak loses 4. All 3 work together. Find the time to fill 160 litres.** ☐ **min**

Working: _____

4 **Parking: $1.50 for the first hour, then $0.60 for each extra 30 minutes (a part counts as a whole one). How much for 2 hours 45 minutes? $** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

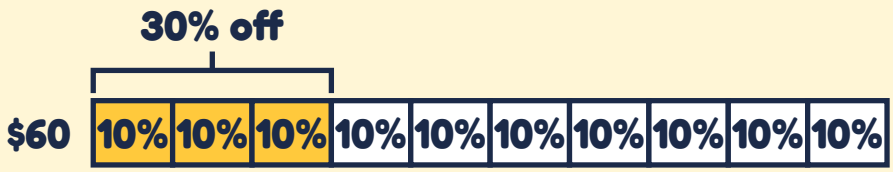
Learn it: Percentage

My goal. I can find a percentage of an amount from 10%, 5% and 1%. I can find a discount and GST, and avoid percentage traps.

Percent means out of 100. Build other percentages from 10%, 5% and 1%. A discount comes off the first cost. GST is charged on the cost after the discount.

Worked example 1

A bag costs $60. It has 30% off. GST of 9% is charged on the cost after the discount. How much does Ali pay?



1. 10% of $60 = $6. So 30% = 3 × $6 = $18 off.
2. The cost after the discount is $60 − $18 = $42.
3. 1% of $42 = $0.42. So 9% = 9 × $0.42 = $3.78 GST.
4. Ali pays $42 + $3.78 = $45.78.

Answer: Ali pays $45.78.

Worked example 2

Find 35% of $160.

1. 10% of $160 = $16.
2. 5% is half of 10%, so 5% = $8.
3. 35% = 10% + 10% + 10% + 5% = $16 + $16 + $16 + $8 = $56.

Answer: 35% of $160 is $56.

How to

1. Find 10%. Build the percentage you need.
2. Take off the discount first.
3. Find the GST on the new amount.

Your turn

Answer at the back of the book.

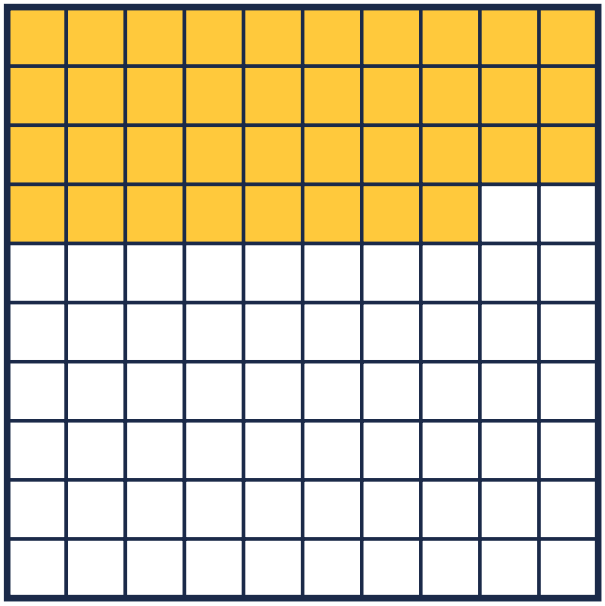
Try **A jacket costs $75. It has 20% off. GST of 9% is charged on the reduced cost.**

How much does Mei pay? $ ☐

Start here

Do these first. Take your time.

1 **The grid has 100 small squares. Some squares are shaded.**



(a) The percentage shaded is ☐ %.

(b) The percentage not shaded is ☐ %.

2 **Find each amount. Use 10%, 5% and 1%.**

(a) 10% of $340 = $ ☐

(b) 5% of $340 = $ ☐

(c) 15% of $340 = $ ☐

3 **Write each as a percentage.**

(a) 18 out of 25 pupils wear glasses. That is ☐ %.

(b) 7 out of 20 pupils cycle. That is ☐ %.

4 **A bag costs $120. It has 15% off.**

(a) 10% of $120 = $ ☐

(b) The discount is $ ☐

(c) The bag now costs $ ☐

Start here

Keep going. Check your work.

5 **A watch costs $64 before GST. GST is 9% of the cost before GST.**

(a) The GST is $ ☐

(b) The watch costs $ ☐ with GST.

6 **Ravi has $150. He spends 20% of his money on a game. Then he spends 40% of the money that is left on a book. How much money does Ravi have now? $**

☐

Working: _____

7 **There are 40 children in a class. 35% walk to school. 25% cycle. The rest take the bus.**

(a) ☐ children walk.

(b) ☐ children cycle.

(c) ☐ children take the bus.

8 **Mei scored 21 out of 30 in Quiz 1 and 34 out of 50 in Quiz 2.**

(a) Quiz 1 is ☐ %.

(b) Quiz 2 is ☐ %.

(c) She did better in Quiz 1 Quiz 2

9 **T-shirts cost $20 each. Shop P says buy 2 and get 1 free. Shop Q says 30% off each T-shirt. Siti needs 3 T-shirts.**

(a) At Shop P she pays $ ☐

(b) At Shop Q she pays $ ☐

(c) The cheaper shop is Shop P Shop Q

Start here

Nearly there. You can do it.

10 **A shirt costs $50. The cost goes up by 20%. Then the new cost goes down by 20%.**

(a) After going up, the shirt costs $ ☐

(b) After going down, it costs $ ☐

(c) Is the cost back to $50? Yes No

11 **Each bar is cut into equal parts. Write the percentage that is shaded.**

(a) Bar 1: ☐ %

(b) Bar 2: ☐ %



12 **Circle True or False.**

(a) 50% of 80 is 40. True False

(b) 10% of 90 is 9. True False

(c) A 20% discount and then a 10% discount is the same as a 30% discount.
True False

13 **Aisha has $200. She spends 40% of it on shoes. She spends 25% of the rest on a gift. How much money does she have left? $** ☐

Working: _____

14 **Ali pays 9% GST on a $250 sofa. Then he pays $20 to have it delivered. The delivery has no GST. How much does he pay in all? $** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **A $200 coat has 10% off. Then there is a further 20% off the new cost.**

(a) The coat finally costs $ ☐

(b) The 2 discounts together are ☐ % of $200.

Working: _____

Hint. Take the 20% off the new cost, not off $200.

2 **Beyond Level 5: Working back from a percentage**

After a 20% discount, a bag costs $60. What was the cost before the discount? $ ☐

Working: _____

Hint. $60 is 80% of the old cost. What is 10% of the old cost?

3 **Beyond Level 5: GST in the cost**

A table costs $218 with GST. GST is 9% of the cost before GST. How much GST is in $218? $ ☐

Working: _____

Hint. $218 is 109% of the cost before GST. What is 1%?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 5: Percentage change**

A cost goes up from $20 to $50, and then it goes back down from $50 to $20. The change is always compared with the amount you start from.

(a) The percentage increase from $20 to $50 is ☐ %.

(b) The percentage decrease from $50 to $20 is ☐ %.

Working: _____

_____

_____

Hint. Increase = change out of the starting amount. Which is the start?

2 **In a class of 40 children, 40% are boys. 25% of the boys and 50% of the girls wear glasses. What percentage of the whole class wears glasses?** ☐ %

Working: _____

_____

_____

_____

Hint. Find the number of boys and girls first. Then count glasses.

3 **A jar has 50 beads. 60% of the beads are blue. The rest are red. Blue beads are added to the jar, and no red beads are taken out. Now 80% of the beads are blue. How many blue beads were added?** ☐

Working: _____

_____

_____

_____

Hint. The red beads stay the same. Find their share of the new total.

✔ Check yourself

No hints. Try each one on your own.

1 **Find the amounts.**

(a) 10% of $180 = $ ☐ (b) 5% of $180 = $ ☐

(c) 15% of $180 = $ ☐

Working: _____

2 **A book costs $40. It has 25% off. GST of 9% is charged on the reduced cost. Find the total cost. $** ☐

Working: _____

3 **Kumar has $80. He spends 25% of it on lunch. He spends 50% of the rest on a gift. How much does he have left? $** ☐

Working: _____

4 **Test 1: 33 out of 50. Test 2: 16 out of 25. Circle the test with the higher percentage.** **Test 1** **Test 2**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

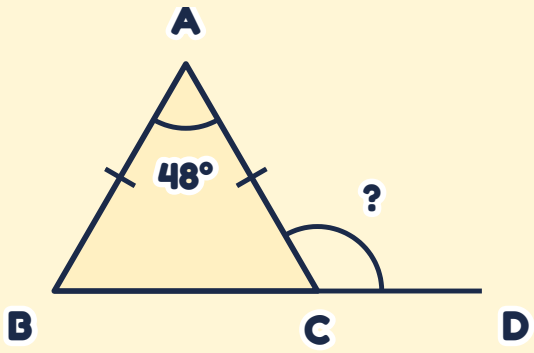
Learn it: Angles, triangles and quadrilaterals

My goal. I can use angle facts to find unknown angles step by step, and give a reason for every step.

Angle facts: angles on a straight line add up to 180°. Angles at a point add up to 360°. Vertically opposite angles are equal. The angles of a triangle add up to 180°. Drawings are not to scale.

Worked example

In the figure, AB = AC and angle BAC = 48°. BCD is a straight line. Find angle ACD. Give a reason for each step.



1. AB = AC, so triangle ABC is isosceles. The angles at B and C are equal.
2. The angles of a triangle add up to 180°. 180° − 48° = 132°.
3. Angle ACB = 132° ÷ 2 = 66° (equal base angles).
4. BCD is a straight line, so ACD = 180° − 66° = 114° (angles on a straight line).

Answer: Angle ACD = 114°.

How to

1. Find one angle. Write the reason.
2. Use it to find the next angle.
3. Check: do the angles add up?

Reasons

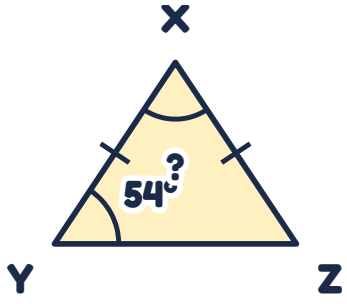
- Straight line: 180°.
- Point: 360°.
- Triangle: 180°.

Your turn

Answer at the back of the book.

Try

In triangle XYZ, XY = XZ and angle Y = 54°. Find angle X. ☐ °



Start here

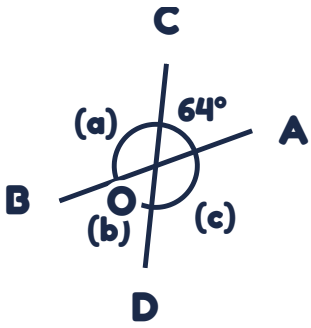
Do these first. Take your time.

1 **Straight lines AB and CD meet at point O. Angle AOC = 64°. Find each angle.**

(a) Angle BOC (a) = ☐ °

(b) Angle BOD (b) = ☐ °

(c) Angle AOD (c) = ☐ °

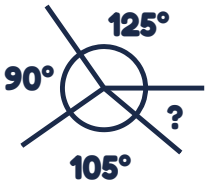


2 **Find the missing angles. The drawings are not to scale.**

(a) On the straight line: ☐ °

(b) At the point: ☐ °

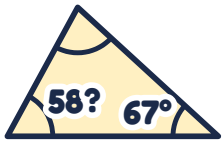


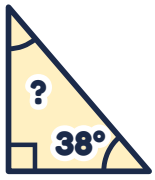


3 **Find the missing angle in each triangle. The second triangle has a right angle.**

(a) Left triangle: ☐ °

(b) Right triangle: ☐ °

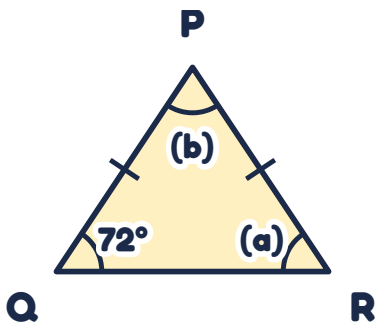




4 **In triangle PQR, PQ = PR. Angle Q = 72°.**

(a) Angle R (a) = ☐ °

(b) Angle P (b) = ☐ °



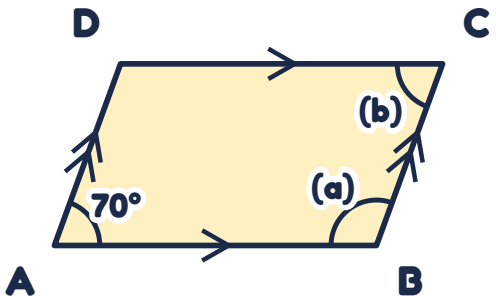
Start here

Keep going. Check your work.

5 ABCD is a parallelogram. Angle A = 70°.

(a) Angle B (a) = ☐ °

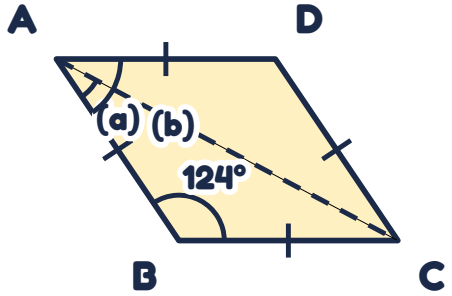
(b) Angle C (b) = ☐ °



6 ABCD is a rhombus. Angle ABC = 124°. The dashed line is the diagonal AC.

(a) Angle BAC (a) = ☐ °

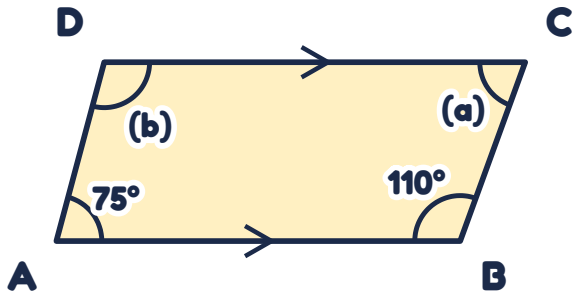
(b) Angle BAD (b) = ☐ °



7 ABCD is a trapezium. AB is parallel to DC. Angle A = 75° and angle B = 110°.

(a) Angle C (a) = ☐ °

(b) Angle D (b) = ☐ °



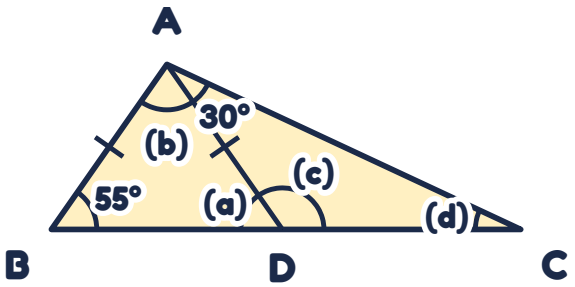
8 BDC is a straight line. AB = AD. Angle ABD = 55° and angle DAC = 30°.

(a) Angle ADB (a) = ☐ °

(b) Angle BAD (b) = ☐ °

(c) Angle ADC (c) = ☐ °

(d) Angle ACD (d) = ☐ °



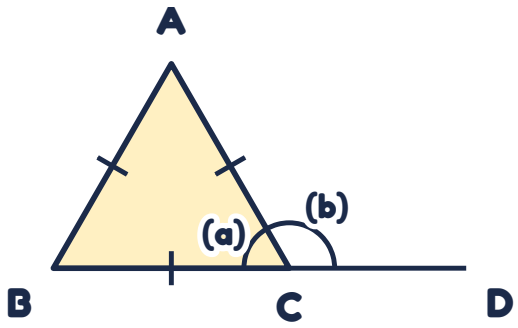
Start here

Nearly there. You can do it.

9 **ABC is an equilateral triangle. BCD is a straight line.**

(a) Angle ACB (a) = ☐ °

(b) Angle ACD (b) = ☐ °



10 **An isosceles triangle has an angle of 70°. The other 2 angles are not known yet.**

(a) If 70° is one of the 2 equal angles, the third angle is ☐ °.

(b) If 70° is the angle between the 2 equal sides, each equal angle is ☐ °.

(c) The number of different triangles is ☐ .

11 **The minute hand of a clock turns through 360° in 60 minutes.**

(a) In 20 minutes it turns through ☐ °.

(b) At 4 o'clock the angle between the hands is ☐ °.

(c) At 7 o'clock the smaller angle between the hands is ☐ °.

12 **Circle the correct reason for each step.**

(a) Angle R = 50° because the
angles on a straight line / angles of a triangle / angles at a point
make 180°. (P = 55°, Q = 75°)

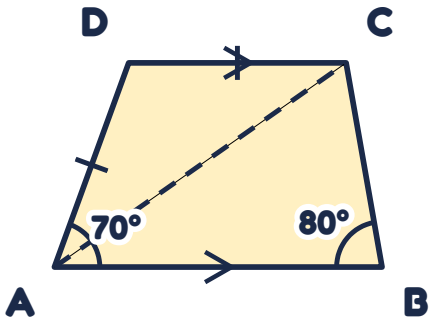
(b) 112° = 180° − 68° because
angles on a straight line / vertically opposite angles / angles at a point
make 180°.

(c) In a parallelogram A = 70° and B = 110° because
opposite angles / neighbouring angles make 180°.

Think harder

Hints are upside down. Try first!

1 **ABCD is a trapezium. AB is parallel to DC. AD = DC. Angle A = 70° and angle B = 80°. The dashed line is the diagonal AC.**



(a) Angle ADC = ☐ °

(b) Angle DCA = ☐ °

(c) Angle ACB = ☐ °

Working: _____

Hint. Find D first. Then use the isosceles triangle ADC.

2 **An isosceles triangle has whole-number angles. Find the angle at the top (between the 2 equal sides) in each case.**

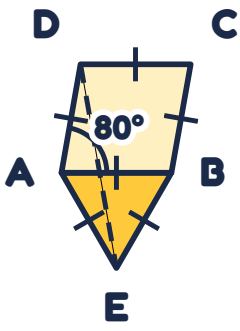
(a) The top angle is 24° more than each equal angle. The top angle is ☐ °.

(b) Each equal angle is 24° more than the top angle. The top angle is ☐ °.

Working: _____

Hint. Make 3 equal parts and the extra 24° (or 2 extra 24°s).

3 **ABCD is a rhombus. ABE is an equilateral triangle. E is on the other side of AB from D. Angle DAB = 80°. Find angle AED.** ☐ °



Working: _____

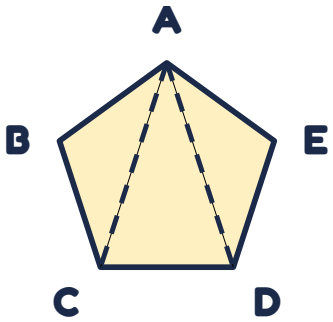
Hint. AD = AB = AE. What kind of triangle is ADE?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 5: Angles of a regular pentagon

The dashed lines split the pentagon into 3 triangles. A regular pentagon has 5 equal sides and 5 equal angles.



(a) The 5 angles of the pentagon add up to ☐°.

(b) Each angle of a regular pentagon is ☐°.

Working: _____

Hint. Each triangle has angles that add up to 180°. Count the triangles.

2 Beyond Level 5: Clock angles

The minute hand turns 6° each minute. The hour hand turns 0.5° each minute. At 4 o'clock the angle between the hands is 120°. Find the smaller angle between the hands at 4:20. ☐°

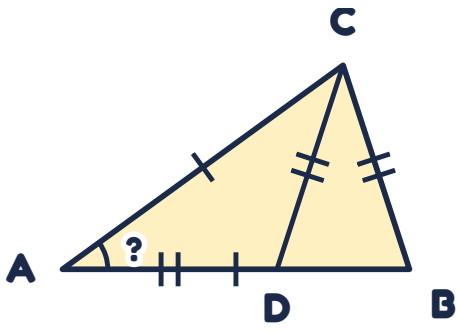
Working: _____

Hint. At 4:20, how far has each hand turned from 12?

3

In triangle ABC, AB = AC. D is on AB so that AD = DC = CB. The marks show equal sides.

Find angle A. ☐°



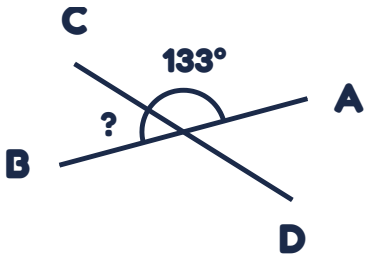
Working: _____

Hint. Call angle A one unit. Use the isosceles triangles one at a time.

✔ Check yourself

No hints. Try each one on your own.

1 Straight lines AB and CD meet at point O. Angle AOC = 133°. Find angle COB. ☐°



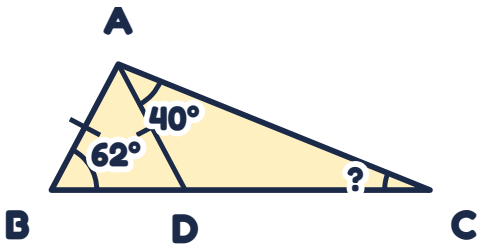
2 In triangle PQR, PQ = PR and angle P = 100°. Find angle Q. ☐°

Working: _____

3 ABCD is a parallelogram. Angle B = 115°. Find angle D. ☐°

Working: _____

4 BDC is a straight line. AB = AD. Angle ABD = 62° and angle DAC = 40°. Find angle ACD. ☐°



My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Patterns and number sense

My goal. I can find the rule of a growing pattern and say it in words. I can use the rule for a far figure and backwards.

Look at what stays the same and what grows each time. Make a table. Say the rule in words. Use the rule for any figure, and backwards from a total.

Worked example

Tiles make a plus sign. Figure 1 has 5 tiles and figure 2 has 9 tiles. Figure 3 has arms of 3 tiles. How many tiles are in figure 10? Which figure has 81 tiles?

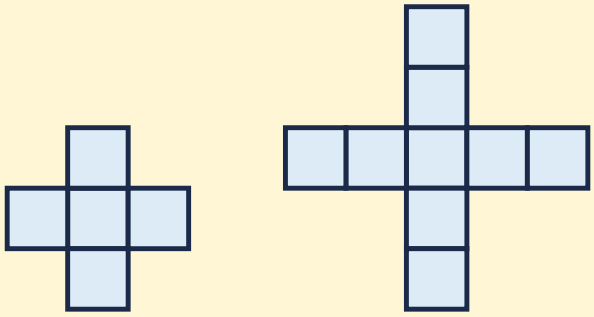


Figure 1 **Figure 2**

1. Each figure has 4 arms and 1 middle tile. In figure n each arm has n tiles.
2. Rule: tiles = 4 × figure number + 1.
3. Figure 10: 4 × 10 + 1 = 41 tiles.
4. Backwards: 81 − 1 = 80, and 80 ÷ 4 = 20. Figure 20 has 81 tiles.

Answer: Figure 10 has 41 tiles. Figure 20 has 81 tiles.

How to

1. Count the figures. Make a table.
2. See what each new figure adds, and why.
3. Say the rule in words. Use it forwards and backwards.

Your turn

Answer at the back of the book.

Try **Matchstick hexagons are joined in a row. Each new hexagon shares one side with the hexagon before it. How many sticks do 12 hexagons need?**



Figure 1 **Figure 2**

Start here

Do these first. Take your time.

1 **Matchstick hexagons are joined in a row. Each new hexagon shares one side with the hexagon before it. Figure 1 has 6 sticks. Figure 2 has 11 sticks.**

(a) 4 hexagons need ☐ sticks.

(b) 5 hexagons need ☐ sticks.

(c) Each new hexagon adds ☐ sticks.



Figure 1

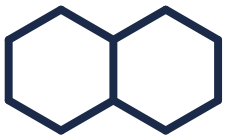


Figure 2

2 **Use the hexagon pattern. Complete the rule and use it.**

(a) sticks = ☐ × hexagons + ☐

(b) 20 hexagons need ☐ sticks.

3 **Mei has some matchsticks. She makes hexagons in a row, as in the pattern.**

(a) 96 sticks make ☐ hexagons.

(b) 100 sticks make at most ☐ hexagons, with ☐ sticks left over.

Working: _____

4 **Look at the pattern: 7, 12, 17, 22, ...**

(a) The next 2 numbers are ☐ and ☐.

(b) The 10th number is ☐.

(c) Is 100 in the pattern? Yes No

Working: _____

Start here

Keep going. Check your work.

5 **Tiles make a plus sign. Figure n has a middle tile and 4 arms of n tiles each.**

(a) Figure 4 has ☐ tiles.

(b) Figure 15 has ☐ tiles.

(c) tiles = ☐ × figure number + ☐

(d) The figure with 81 tiles is figure ☐.

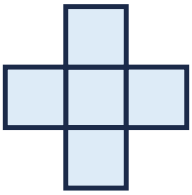


Figure 1

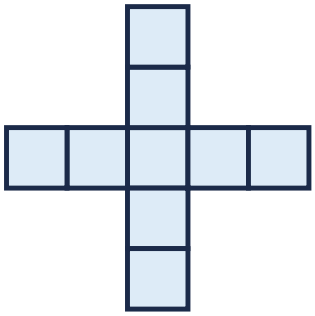


Figure 2

6 **Adult tickets cost $9 each. Child tickets cost $5 each. A group buys 15 tickets and pays $107 altogether. Make a table to guess and check.**

(a) Adult tickets: ☐

(b) Child tickets: ☐

Working: _____

7 **A shop has 17 stools and chairs. A stool has 3 legs and a chair has 4 legs. There are 61 legs altogether.**

(a) There are ☐ chairs.

(b) There are ☐ stools.

Working: _____

8 **A quiz has 12 questions. A correct answer scores 5 marks. A wrong answer loses 2 marks. Siti answers every question and scores 39 marks.**

(a) She gets ☐ answers wrong.

(b) She gets ☐ answers right.

Working: _____

Start here

Nearly there. You can do it.

9 **Long tables are joined end to end in a row. Each table has 2 chairs on each long side. 1 chair sits at each end.**

(a) 5 tables have ☐ chairs.

(b) chairs = ☐ × tables + ☐

(c) 50 chairs belong to ☐ tables.

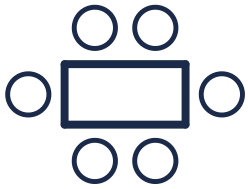


Figure 1

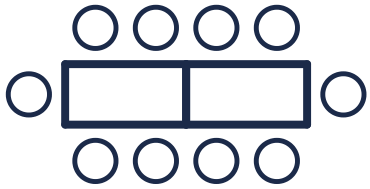


Figure 2

10 **The table shows a pattern.**

Figure	Sticks
1	7
2	10
3	13
4	16

(a) sticks = ☐ × figure number + ☐

(b) Figure 12 has ☐ sticks.

(c) The figure with 100 sticks is figure ☐.

11 **Circle True or False.**

(a) The numbers 5, 9, 13, 17, ... include 101. True False

(b) 10 hexagons need 51 sticks, so 20 hexagons need 102 sticks.
True False

(c) No number in the pattern 6, 11, 16, 21, ... ends in 3. True False

Think harder

Hints are upside down. Try first!

1 **A purse has 24 coins. Some are 20¢ coins and the rest are 50¢ coins. The coins are worth $9.60 altogether. How many 50¢ coins are there?** ☐

Working: _____

_____

_____

Hint. Suppose all are 20¢ coins. How many cents short is that?

2 **Tiles make staircases. Figure 1 has 1 tile, figure 2 has 3 tiles, figure 3 has 6 tiles and figure 4 has 10 tiles.**

(a) Figure 10 has ☐ tiles.

(b) The figure with 105 tiles is figure ☐.

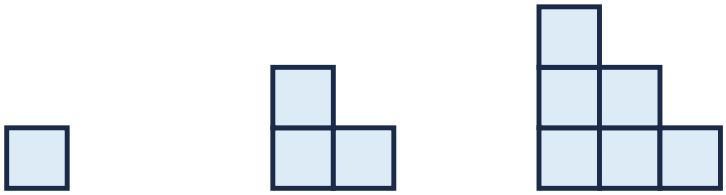


Figure 1 **Figure 2** **Figure 3**

Working: _____

Hint. Figure n adds a column of n tiles. Two staircases make a rectangle.

3 **Look at the squares. Each new square adds an L-shaped layer of tiles (shaded). Find 1 + 3 + 5 + 7 + … + 49. The numbers go up by 2 each time.** ☐

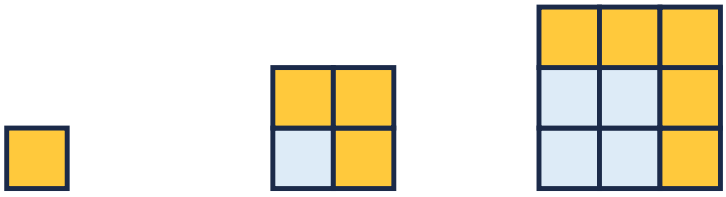


Figure 1 **Figure 2** **Figure 3**

Working: _____

Hint. 1, 1 + 3, 1 + 3 + 5: what kind of numbers are they?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 5: Adding a pattern

Add 4 + 7 + 10 + 13 + … + 100. The numbers go up by 3 each time. ☐

Working:

Hint. How many numbers are there? Pair the first with the last.

2 Beyond Level 5: Handshakes

Every person in a group shakes hands once with every other person.

(a) 5 people make ☐ handshakes.

(b) 12 people make ☐ handshakes.

Working:

Hint. n people each shake n - 1 hands. Each handshake counts twice.

3

A quiz has 30 questions. Each right answer scores 4 marks. A wrong answer loses 1 mark. A blank scores 0. Ali leaves 3 questions blank and scores 73 marks.

How many questions does he get right? ☐

Working:

Hint. Only answered questions count. Suppose all of them are right.

✔ Check yourself

No hints. Try each one on your own.

1 **The numbers go 6, 11, 16, 21, … What is the 15th number?** ☐

Working: _____

2 **The rule for a pattern is: dots = 3 × figure number + 2. Which figure has 41 dots?** ☐

Working: _____

3 **A box holds 20 stools and chairs. A stool has 3 legs and a chair has 4 legs. There are 70 legs altogether. How many chairs are there?** ☐

Working: _____

4 **Matchstick hexagons are joined in a row. The rule is sticks = 5 × hexagons + 1. Mei has 130 sticks. What is the most hexagons she can make?** ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Mixed heuristics

My goal. I can choose a way to start a puzzle, and check my answer against every clue.

Choose a start: draw a model, make a list or a table, work backwards, guess and check, or use a logic grid. Then check your answer against every clue.

Worked example

Ali, Mei and Siti each have a different pet: a dog, a cat or a fish. Ali does not have a fish. Mei does not have a cat or a fish. Who has the cat?

	Dog	Cat	Fish
Ali	No	Yes	No
Mei	Yes	No	No
Siti	No	No	Yes

1. Mei does not have a cat or a fish, so Mei has the dog. Write Yes for Mei and dog.
2. The dog is taken. Write No in the dog column for Ali and Siti, and No in the other columns for Mei.
3. Ali has no fish and no dog, so Ali has the cat.
4. Check: Siti has the fish. Every clue is true.

Answer: Ali has the cat.

How to

1. Use a clue that tells you something for sure.
2. A Yes means No in the rest of its row and column.
3. Check every clue at the end.

Ways to start

- Draw a model.
- List in order or make a table.
- Work backwards.
- Guess and check.

Your turn

Answer at the back of the book.

Try **Ben, Kumar and Aisha each choose a different drink: tea, juice or milk. Ben does not choose juice or milk. Aisha does not choose tea. Kumar does not choose milk. Who chooses juice?** ☐

Start here

Do these first. Take your time.

1 **Mei picks a number. She multiplies it by 3, adds 8 and then halves the result. She gets 19. What number did Mei pick?** ☐

Working: _____

_____

2 **Kumar spends 1/4 of his money. Then he spends $6. He has $15 left. How much money did Kumar have at first? $** ☐

Working: _____

3 **Aisha, Wei and Ravi own a turtle, a parrot and a goldfish, one each. Ravi owns neither the parrot nor the goldfish. Aisha does not own the goldfish. Who owns the parrot?** ☐

4 **Two numbers add up to 52. The larger number is 14 more than the smaller number.**

(a) The larger number is ☐.

(b) The smaller number is ☐.

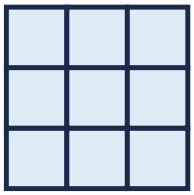
5 **Count the squares of every size in a 3 by 3 grid.**

(a) Squares 1 by 1: ☐

(b) Squares 2 by 2: ☐

(c) Squares 3 by 3: ☐

(d) All the squares: ☐



Start here

Keep going. Check your work.

6 **Exactly one of 3 children broke a vase. Exactly one of the 3 children is telling the truth.**
Ali says, "I did not break the vase."
Ben says, "Ali broke the vase."
Siti says, "Ben is lying." Who broke the vase? ☐

Working: _____

7 **Dad is 40 years old. Ali is 8 years old. In how many years will Dad be 3 times as old as Ali?** ☐ **years**

Working: _____

8 **Ben gives 1/3 of his stickers to Ali. He gives 1/2 of the rest to Mei. 12 stickers are left. Find the number of stickers Ben began with.** ☐

Working: _____

9 **A farm has hens and cows, 20 animals in all. They have 56 legs altogether. Hens have 2 legs each and cows have 4 legs each.**

(a) There are ☐ cows.

(b) There are ☐ hens.

Working: _____

Start here

Nearly there. You can do it.

10 **The product of 2 whole numbers is 72. Their sum is 17. List the pairs with product 72 to find them.**

(a) The numbers are ☐ and ☐.

Working: _____

11 **Use the digit cards 1, 2 and 3 once each to make 3-digit numbers.**



(a) The number of different numbers is ☐.

(b) The even numbers are ☐ and ☐.

(c) The number closest to 250 is ☐.

Working: _____

12 **Ali writes the whole numbers from 1 to 60. How many times does he write the digit 5?** ☐

Working: _____

Think harder

Hints are upside down. Try first!

1 **Ben, Kumar, Aisha and Wei each play a different sport: swimming, tennis, football or badminton. Write Yes or No in the grid. Then write the sport for each child.**
Clue 1: Ben does not play tennis or badminton.
Clue 2: Kumar does not play swimming or tennis.
Clue 3: Wei does not play swimming, badminton or tennis.

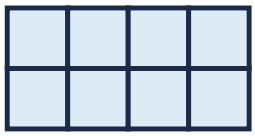
	swimming	tennis	football	badminton
Ben				
Kumar				
Aisha				
Wei				

(a) Ben plays ☐ (b) Kumar plays ☐

(c) Aisha plays ☐ (d) Wei plays ☐

Hint. Which child has the most clues? Start with Wei.

2 **Count the squares and the rectangles in a 2 by 4 grid. A square also counts as a rectangle.**



(a) The number of squares (all sizes) is ☐.

(b) The number of rectangles (all sizes) is ☐.

Working: _____

Hint. Count by size: 1 by 1, 2 by 1, 3 by 1, 4 by 1 … and then 2 high.

3 **In a cafe, 1/4 of the customers leave. Then 3 more customers arrive. Then 1/3 of the customers in the cafe leave. 14 customers are left. How many customers were in the cafe at first?** ☐

Working: _____

Hint. Undo the last step first. 14 is what part of the number before?

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **For how many 3-digit numbers do the digits sum to 7? Make an ordered list to be sure.** ☐

Working: _____

_____

_____

Hint. Sort by the first digit: 1__, 2__, … How many for each?

2 **Ali has 24 coins. They are 10¢, 20¢ and 50¢ coins. The number of 10¢ coins equals the number of 50¢ coins. The coins are worth $7.00 in all.**

(a) There are ☐ coins of 50¢.

(b) There are ☐ coins of 10¢.

(c) There are ☐ coins of 20¢.

Working: _____

_____

_____

Hint. Join a 10¢ coin and a 50¢ coin as a pair. How much is a pair?

3 **Beyond Level 5: The worst case**

A drawer has 10 red socks, 10 blue socks and 10 black socks. You take socks out in the dark. How many socks must you take to be sure of 2 socks of the same colour? ☐

Working: _____

_____

_____

Hint. Think of the worst luck. How many socks can you take with no pair?

✔ Check yourself

No hints. Try each one on your own.

1 **Siti picks a number. She adds 7 and then multiplies by 4. She gets 60. What number did Siti pick?** ☐

Working: _____

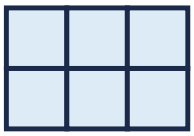
2 **Mei spends 2/5 of her money. Then she spends $9. She has $12 left. How much money did Mei have at first? $** ☐

Working: _____

3 **A box holds 9 hens and goats. They have 26 legs altogether. A hen has 2 legs and a goat has 4 legs. How many goats are there?** ☐

Working: _____

4 **Count the squares of every size in a 2 by 3 grid.** ☐



My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Million puzzles	☐	☐	☐	☐	/ 4	
2	Fraction feats	☐	☐	☐	☐	/ 4	
3	Before and after	☐	☐	☐	☐	/ 4	
4	Triangle areas	☐	☐	☐	☐	/ 4	
5	Volume vault	☐	☐	☐	☐	/ 4	
6	Rate racers	☐	☐	☐	☐	/ 4	
7	Percentage puzzles	☐	☐	☐	☐	/ 4	
8	Angle chasers	☐	☐	☐	☐	/ 4	
9	Pattern generalisers	☐	☐	☐	☐	/ 4	
10	Challenge day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 5.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Million puzzles

Your turn
page 6

Answer: 7 964 310
Why: Just below 8 000 000: start with 7, then the greatest order: 7 964 310 (35 690 away). Just above: start with 9, then the smallest order: 9 013 467 (1 013 467 away). 7 964 310 is closer.

START HERE

1
page 7

Answer: (a) 6 205 030 (b) 70 000 (c) 8604
Why: (a) Read the groups: 6 million, 205 thousand, 30. The hundreds place of the last group is empty, so write 6 205 030. (b) The 7 is in the ten thousands place, so it stands for 70 000. (c) 8 604 000 ÷ 1000 = 8604.

2
page 7

Answer: 940 621, 4 096 213, 4 096 231, 4 906 213
Why: 940 621 has only 6 digits, so it is the smallest. The other three start 4 million. Compare the next digits: 0 comes before 9, so 4 906 213 is last. Then 213 is less than 231.

3
page 7

Answer: (a) 7 850 000 (b) 7 800 000 (c) 8 000 000
Why: Look at the digit just after the place you round to. (a) 849 520: the hundreds digit is 5, round up to 850 000. (b) 49 520 is less than 50 000, round down to 7 800 000. (c) 849 520 is more than 500 000, round up to 8 000 000.

4
page 7

Answer: (a) 9 875 410 (b) 1 045 789
Why: (a) A multiple of 5 ends in 0 or 5. Ending in 0 gives 9 875 410. Ending in 5 gives only 9 874 105, which is less. (b) The first digit cannot be 0, so start with 1. Then 0, 4, 5, 7, 8, 9 in order: 1 045 789.

5
page 7

Answer: 6 632 315
Why: 2 hundred thousands is 200 000, and 5 ten thousands is 50 000. 6 482 315 + 200 000 = 6 682 315. Then 6 682 315 - 50 000 = 6 632 315. Only the hundred thousands and ten thousands digits change.

6
page 8

Answer: (a) must be wrong (b) could be right (c) could be right (d) must be wrong
Why: (a) 4000 × 50 = 200 000, but the answer shown is only about 20 000. (b) 4 000 000 + 3 000 000 = 7 000 000, and 6 912 005 is close. (c) 8000 × 300 = 2 400 000, close to the answer. (d) 6000 ÷ 15 = 400, but 4050 is ten times too big.

7
page 8

Answer: (a) (9 - 2) × 3 + 1 = 22 (b) 9 - 2 × (3 + 1) = 1 (c) 2 400 000 ÷ (300 + 100) × 2 = 12 000
Why: Brackets first. (a) (9 - 2) × 3 + 1 = 7 × 3 + 1 = 22. (b) 9 - 2 × (3 + 1) = 9 - 2 × 4 = 9 - 8 = 1. (c) 300 + 100 = 400. Then 2 400 000 ÷ 400 = 6000, and 6000 × 2 = 12 000.

8
page 8

Answer: 104 cartons
Why: 1 250 000 ÷ 500 = 2500 boxes. Then 2500 ÷ 24 = 104 remainder 4, since 24 × 104 = 2496. Only full cartons count, so 104.

9
page 8

Answer: (a) 8650 (b) 8749
Why: A number rounds to 8700 when it is from halfway below to just before halfway above. Halfway marks are 8650 and 8750. 8650 rounds up to 8700, so it counts. 8750 rounds up to 8800, so the greatest is 8749.

10
page 8

Answer: (a) True (b) False (c) True
Why: (a) 54 × 100 000 = 5 400 000, so it is true. (b) 1 000 000 - 1 = 999 999, which has 6 nines, so it is false. (c) The greatest 6-digit number is 999 999 and the smallest 7-digit number is 1 000 000, so it is true.

11
page 9

Answer: (a) Dune (b) 395 550
Why: (a) All four numbers have 7 digits. Compare the hundred thousands: Dune has 4, the others 2 or 0. Dune is the greatest. (b) Cedar is smallest. 1 420 050 - 1 024 500 = 395 550.

12
page 9

Answer: 472 630
Why: Clue 2: the number is from 465 000 to 474 999. With Clue 3 the thousands digit is 2, so it starts 472 (465 and 474 do not fit). Clue 5: ones 0. Clue 4: the tens and hundreds digits are 3 and 6, or 1 and 2 (2 is already used), or 4 and 8 (4 is used). So 472 630. Check all digits differ: 4, 7, 2, 6, 3, 0.

13
page 9

Answer: 14 250 people
Why: Monday: 3100. Tuesday: 2 × 3100 = 6200. Wednesday: 6200 - 1250 = 4950. Total: 3100 + 6200 + 4950 = 14 250. The 48 000 seats are not needed.

14
page 9

Answer: 4 500 000 and 5 499 999
Why: Numbers from 4 500 000 up to 5 499 999 round to 5 000 000. 4 499 999 is just below, so it rounds to 4 000 000. 5 500 000 rounds up to 6 000 000.

THINK HARDER

1
page 10

Answer: 6 435 708
Why: Clue 2: from 6 350 000 to 6 449 999, so it starts 6 3 5 or more, or 6 4. Clue 3: the ten-thousands digit is 1 less than the hundred-thousands digit. 6 3 5 does not fit (5 is not 2). So it starts 6 4 3. Digits so far: 6, 4, 3 and 0 (the tens). Clue 6: the other three digits add to 33 - 6 - 4 - 3 - 0 = 20. The thousands digit t and the hundreds digit t + 2: t + (t + 2) + ones = 20. Try t = 5: 5 + 7 = 12, ones = 8. All different: 6, 4, 3, 5, 7, 0, 8. So 6 435 708.

2
page 10

Answer: (a) 10
Why: From 195 000 to 199 999 the digits 1 + 9 are already 10, too big. So the number is 20 _ _ _ _ with the other four digits adding to 2. The thousands digit is at most 4. The ways: 2 in one place (4 ways: 200 002, 200 020, 200 200, 202 000) or 1 and 1 in two places (6 ways: 200 011, 200 101, 200 110, 201 001, 201 010, 201 100). Total 10.

3
page 10

Answer: 69 936
Why: Put the greatest digits first: 9 and 7 lead the two numbers. Try 9__ × 7_ and 7__ × 9_ with the other digits 5, 3, 2 in order. 752 × 93 = 69 936 beats 953 × 72 = 68 616 and 732 × 95 = 69 540. The best is 752 × 93 = 69 936.

GO FURTHER

1
page 11

Answer: 544 320
Why: The first digit has 9 choices (1 to 9). The second digit can be any digit except the first: 9 choices (0 is allowed now). The third has 8, then 7, 6, 5, 4. Multiply: 9 × 9 × 8 × 7 × 6 × 5 × 4 = 544 320.

2
page 11

Answer: 13 500
Why: Write every number with 3 digits, so 7 becomes 007. Adding extra zeros does not change the total. The numbers 000 to 999 are 1000 numbers. In each of the 3 places, every digit 0 to 9 appears 100 times. One place adds (0 + 1 + ... + 9) × 100 = 4500. Three places add 3 × 4500 = 13 500.

3
page 11

Answer: 101 (3 × 4 × 9 - 7)
Why: 3 × 4 × 9 = 108. Then 108 - 7 = 101, which is 1 away from 100. Trying every way, 99 and 100 cannot be made, and 102 is 2 away (9 × (4 + 7) + 3). So the closest is 101.

4
page 11

Answer: 53 days
Why: In 1 day it prints 2400 × 8 = 19 200 sheets. 1 000 000 ÷ 19 200 = 52.08... days. After 52 days it has printed 998 400 sheets, which is not enough. So it needs 53 days.

CHECK YOURSELF

1
page 12

Answer: (a) 200 000 (b) 5 300 000
Why: (a) The 2 is in the hundred thousands place: 200 000. (b) 5 280 436: the ten thousands digit is 8, so round up to 5 300 000.

2
page 12

Answer: 3 976 520
Why: Just below: start with 3, then greatest order: 3 976 520 (23 480 away). Just above: start with 5, then smallest order: 5 023 679 (1 023 679 away). The number just below is much closer.

3
page 12

Answer: (8 + 12) ÷ 4 × 2 = 10
Why: (8 + 12) ÷ 4 × 2 = 20 ÷ 4 × 2 = 5 × 2 = 10. Without brackets the line gives 8 + 3 × 2 = 14.

4
page 12

Answer: 270 000 passengers
Why: In 1 week: 18 × 1250 = 22 500. In 12 weeks: 22 500 × 12 = 270 000.

Unit 2 · Fraction feats

Your turn
page 13

Answer: 30 beads
Why: 1/4 of 120 = 30, so 90 beads are left. 2/3 of 90 = 60 beads for the necklace. 90 - 60 = 30 beads are left. (Or: 1/3 of the 90 is left, which is 30.)

START HERE

1
page 14

Answer: (a) 7/10 (b) 3/8 (c) 2 1/4
Why: Sharing means dividing. a ÷ b = a/b. (a) 7 ÷ 10 = 7/10. (b) 3 ÷ 8 = 3/8. (c) 9 ÷ 4 = 9/4. 4 goes into 9 two times with 1 left, so 2 and 1/4.

2 page 14
Answer: (a) 5/6 (b) 4/7 (c) 2/3 of 24
Why: (a) 4/5 is 1/5 short of 1, and 5/6 is 1/6 short of 1. 1/6 is less than 1/5, so 5/6 is greater. (b) 5/9 = 35/63 and 4/7 = 36/63, so 4/7 is greater. (c) 3/4 of 20 = 15 and 2/3 of 24 = 16.

3 page 14
Answer: (a) 24 (b) 40 (c) 16
Why: 1 part is 64 ÷ 8 = 8 cm. (a) 3 parts = 24 cm. (b) 5 parts = 40 cm. (c) 1/4 is 2 parts of the 8, which is 16 cm.

4 page 14
Answer: (a) 11/12 (b) 11/24 (c) 7/12
Why: Change to the same bottom number. (a) 8/12 + 3/12 = 11/12. (b) 20/24 - 9/24 = 11/24. (c) 9/12 - 2/12 = 7/12.

5 page 15
Answer: 4 1/2 km
Why: 6 × 3/4 = 18/4. Divide 18 by 4: 4 with 2 left over, so 4 and 2/4, which is 4 1/2. Or: 3/4 of 6 is 4 1/2.

6 page 15
Answer: 12 cards
Why: 1/3 of 72 = 24, so 72 - 24 = 48 are left. 3/4 of 48 = 36 go to Ben. 48 - 36 = 12. (Or: 1/4 of 48 is left, and 48 ÷ 4 = 12.)

7 page 15
Answer: 1/4 (7/12 = 1/3 + 1/4)
Why: 1/3 = 4/12. 7/12 - 4/12 = 3/12 = 1/4. Both 1/3 and 1/4 have 1 on top and they are different, so 7/12 = 1/3 + 1/4.

8 page 15
Answer: (a) 5/11, 6/13 (b) 10/21 (c) 13
Why: The top number is the position. The bottom number goes up by 2 each time: 2n + 1 for position n. (a) 5/11 and 6/13. (b) 2 × 10 + 1 = 21, so 10/21. (c) 13/27 has 13 on top, and 2 × 13 + 1 = 27, so it is the 13th.

9 page 15
Answer: 7/20 of the book
Why: 2/5 + 1/4 = 8/20 + 5/20 = 13/20 is read. The whole book is 20/20. 20/20 - 13/20 = 7/20.

10 page 15
Answer: (a) 1 1/4 (b) 1 3/4
Why: The line from 0 to 1 is cut into 4 equal parts, so each mark is 1/4. A is 1 whole and 1 more quarter: 1 1/4. B is 1 whole and 3 quarters: 1 3/4.

11 page 16
Answer: (a) 1/4 (b) 2/5 (c) 1 1/3
Why: Multiply the tops and the bottoms, then simplify. (a) 6/24 = 1/4. (b) 6/15 = 2/5. (c) 12/9 = 4/3 = 1 1/3.

12 page 16
Answer: (a) 18 (b) 54
Why: The book is 1 part. The rest is 2 parts, and lunch is half of the rest, which is 1 part. The $18 left is also 1 part. So lunch cost $18 and 3 parts = 3 × $18 = $54.

13 page 16
Answer: 15 children
Why: 3/8 of 40 = 15 walk. 1/4 of 40 = 10 take the bus. 15 + 10 = 25, so 40 - 25 = 15 cycle.

14 page 16

Answer: (a) False (b) True (c) True
Why: (a) 3 ÷ 5 = 3/5, not 5/3. (b) 7/8 = 21/24 and 5/6 = 20/24, so it is true. (c) 1/4 × 1/2 = 1/8, so it is true.

THINK HARDER

1 page 17

Answer: 1/3 + 1/15
Why: Test 1/3: 2/5 - 1/3 = 6/15 - 5/15 = 1/15. That is a unit fraction, so 2/5 = 1/3 + 1/15. Test 1/4: 2/5 - 1/4 = 3/20, not a unit fraction. 1/5 is half of 2/5, so we stop (the 2 parts would be equal). So 1/3 + 1/15 is the only way.

2 page 17

Answer: 120 eggs
Why: After the morning 2/3 of the eggs are left. In the afternoon 1/4 of those are sold, so 3/4 of 2/3 = 1/2 are left. In the evening 2/5 of those are sold, so 3/5 of 1/2 = 3/10 are left. 36 eggs are 3/10 of the whole. 1/10 is 12 eggs, so the whole is 120 eggs.

3 page 17

Answer: (a) 5 1/6 (b) 29/35
Why: Greatest: put the greatest card on top with the smallest card below: 7/2 + 5/3 = 21/6 + 10/6 = 31/6 = 5 1/6. Smallest: put the small cards on top and the big cards below: 2/5 + 3/7 = 14/35 + 15/35 = 29/35. Testing every way shows no other sum is greater or smaller.

GO FURTHER

1 page 18

Answer: 1/12 of the cake
Why: Cut every sixth into 2 equal parts. Then 5/6 becomes 10 parts, and each part is 1/12 of the cake. So 5/6 ÷ 10 = 1/12. Check: 10 × 1/12 = 10/12 = 5/6.

2 page 18

Answer: 4 ways
Why: Test 1/7: 1/6 - 1/7 = 1/42, so 1/7 + 1/42. Test 1/8: 1/6 - 1/8 = 1/24, so 1/8 + 1/24. Test 1/9: 1/6 - 1/9 = 1/18, so 1/9 + 1/18. Test 1/10: 1/6 - 1/10 = 1/15, so 1/10 + 1/15. Test 1/11: 1/6 - 1/11 = 5/66, not a unit fraction. 1/12 is half of 1/6, so stop. There are 4 ways.

3 page 18

Answer: (a) 64 (b) 3
Why: Work backwards. Before the 3rd bounce it was 27 ÷ 3/4 = 36 cm. Before that 36 ÷ 3/4 = 48 cm. Before that 48 ÷ 3/4 = 64 cm. Heights: 64, 48, 36, 27. Half of 64 is 32. 36 is more than 32, and 27 is less, so it is bounce 3.

4 page 18

Answer: 4/9
Why: 1/3 = 1/2 × (1 - 1/3), 1/15 = 1/2 × (1/3 - 1/5), 1/35 = 1/2 × (1/5 - 1/7) and 1/63 = 1/2 × (1/7 - 1/9). In the sum the middle parts cancel, leaving 1/2 × (1 - 1/9) = 1/2 × 8/9 = 4/9. Check by adding: 1/3 + 1/15 = 2/5, + 1/35 = 3/7, + 1/63 = 28/63 = 4/9.

CHECK YOURSELF

1 page 19

Answer: (a) 5/6 (b) 2 1/3
Why: (a) 5 ÷ 6 = 5/6. (b) 7 ÷ 3: 3 goes into 7 two times with 1 left, so 2 and 1/3.

2 page 19
Answer: (a) 13/24 (b) 7/10
Why: (a) 21/24 - 8/24 = 13/24. (b) 4/10 + 3/10 = 7/10.

3 page 19
Answer: 60 marbles
Why: 1/5 of 120 = 24, so 96 are left. 3/8 of 96 = 36 go to Wei. 96 - 36 = 60.

4 page 19
Answer: 1/12 of the bag
Why: 3/4 + 1/6 = 9/12 + 2/12 = 11/12 is used. 12/12 - 11/12 = 1/12 is left.

Unit 3 · Before and after

Your turn page 20
Answer: 25 stickers
Why: The total stays the same: 70 + 50 = 120. After, Jason has 3 units and Amelia has 5 units. 8 units = 120, so 1 unit = 15. Jason now has 3 × 15 = 45. Jason gives 70 - 45 = 25 stickers. Check: Amelia has 50 + 25 = 75 = 5 × 15.

START HERE

1 page 21
Answer: (a) 20 (b) 20 (c) 40
Why: (a) 1/4 of 80 = 20. (b) 80 - 20 = 60 is left. 1/3 of 60 = 20. (c) 60 - 20 = 40 is left.

2 page 21
Answer: (a) 17 (b) 19 (c) 36
Why: (a) 22 - 5 = 17. (b) 14 + 5 = 19. (c) 22 + 14 = 36, and 17 + 19 = 36. Giving does not change the total.

3 page 21
Answer: (a) 18, 47 (b) 29, 29
Why: (a) 12 + 6 = 18 and 41 + 6 = 47. (b) 41 - 12 = 29 and 47 - 18 = 29. When both get older by the same number of years, the difference stays the same.

4 page 21
Answer: 7 stamps
Why: The total stays 38 + 24 = 62. Each has 62 ÷ 2 = 31. Ali has 38 - 31 = 7 stamps too many, so he gives 7. Check: Mei has 24 + 7 = 31.

5 page 21
Answer: 8 children
Why: 1/12 of 48 = 4. 5/12 of 48 = 20 wear glasses. 48 - 20 = 28 do not. 28 - 20 = 8 more.

6 page 22
Answer: (a) green (b) 5/8 (c) 1/2
Why: (a) Only yellow beads were added, so the 15 green beads did not change. (b) At first there are 24 beads: 15/24 = 5/8. (c) Now there are 30 beads: 15/30 = 1/2. The green beads stay the same but they are a smaller part of the bag.

7 page 22
Answer: (a) 15 (b) 75
Why: The bag takes 2 of the 5 parts. The other 3 parts are $45, so 1 part = 45 ÷ 3 = 15. All 5 parts = 5 × 15 = 75.

8 page 22
Answer: (a) 21 (b) 7
Why: The book is 3 units and the pen is 1 unit. 4 units = 28, so 1 unit = 7. The pen costs 7. The book costs 3 × 7 = 21.

9 page 22
Answer: (a) 32 (b) 24
Why: Siti has 4 units and Mei has 3 units. 7 units = 56, so 1 unit = 8. Siti has 4 × 8 = 32. Mei has 3 × 8 = 24.

10 page 23
Answer: (a) the total (b) the difference (c) the blue beads
Why: (a) Ben gains what Ali loses, so the total is the same. (b) Both gain $5, so the gap between them is the same, but the total goes up by $10. (c) Only red beads are put in, so the number of blue beads does not change.

11 page 23
Answer: (a) 60 (b) 36 (c) 24
Why: (a) 45 + 15 = 60. (b) 21 + 15 = 36. (c) 60 - 36 = 24. Before the gift 45 - 21 = 24 too. The difference did not change.

12 page 23
Answer: $60
Why: The shirt is 1 unit and the rest is 3 units. The shoes are 1/3 of the rest, which is 1 unit. So 4 - 1 - 1 = 2 units are left. 2 units = $30, so 1 unit = $15. At first she had 4 × 15 = $60.

13 page 23
Answer: 750 ml
Why: 5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3 of the jug was poured out. 1/3 is 250 ml, so a full jug is 3 × 250 = 750 ml.

14 page 23
Answer: (a) False (b) True (c) True
Why: (a) False: what Ali loses Ben gains. (b) True: both amounts go up by 5. (c) True: if Mei has 1 unit, Ali has 2 units.

THINK HARDER

1 page 24
Answer: $150
Why: 2/5 = 6/15, so Monday takes 6 of 15 units. The rest is 9 units. Tuesday is 1/3 of 9 = 3 units. In all 6 + 3 = 9 units are spent. 9 units = $90, so 1 unit = $10. At first Siti had 15 × 10 = $150.

2 page 24
Answer: 55 beads
Why: After: Siti 3 units, Mei 5 units. 8 units = 120, so 1 unit = 15. Siti has 45 now. She gave away 10, so at first she had 45 + 10 = 55. Check: Mei had 75 - 10 = 65 and 55 + 65 = 120.

3 page 24
Answer: 4 years
Why: The difference is 38 - 10 = 28 years, and it stays 28. When the son is 1 unit the father is 3 units, so the difference is 2 units. 2 units = 28, so 1 unit = 14. The son is 14, which is 4 years from now. Check: father 42 = 3 × 14.

GO FURTHER

1 page 25
Answer: (a) 30 (b) 40 (c) 80
Why: Let Cara have 1 unit. Ben has 1 unit + 10. Ali has twice Ben: 2 units + 20. Altogether: 4 units + 30 = 150. So 4 units = 120 and 1 unit = 30. Cara 30, Ben 40, Ali 80. Check: 30 + 40 + 80 = 150.

2
page 25

Answer: 150 people
Why: The number of people stays the same. At first children are 2/5 of the total. Now they are 1/2. 1/2 - 2/5 = 1/10 of the total is 15 children. So the total is 10 × 15 = 150. Check: children 60 at first, 75 now = half of 150.

3
page 25

Answer: 168 beads
Why: The white beads did not change. At first white is 5/7 of the beads. Now white is 5/6 of the beads. Let the white beads be 5 parts: at first the total is 7 parts, now it is 6 parts. 1 part = 24 beads, because 24 red beads were taken out. At first 7 × 24 = 168 beads. Check: red 48, then 24 of 144 beads is 1/6.

4
page 25

Answer: 45 marbles
Why: Let Ali have 3 units at first. He gives 1 unit and keeps 2 units. Now Ben has 3 times as many: 6 units. After, the total is 2 + 6 = 8 units = 120, so 1 unit = 15. Ali had 3 × 15 = 45. Ben had 6 units - 1 unit = 5 units = 75 at first. Check: 45 + 75 = 120.

CHECK YOURSELF

1
page 26

Answer: $60
Why: The book is 1 unit and the rest is 4 units. The bag is 1/2 of 4 = 2 units. 5 - 1 - 2 = 2 units are left. 2 units = $24, so 1 unit = $12. At first Jenny had 5 × 12 = $60.

2
page 26

Answer: 8 cards
Why: The total is 76, so each has 38. Jamie has 46 - 38 = 8 too many. Jamie gives 8 cards.

3
page 26

Answer: $30
Why: The difference stays $20. After the gift Ray has 3 units and Amelia has 4 units, so the difference is 1 unit. 1 unit = $20. Amelia has $80 now, so the gift is 80 - 50 = $30. Check: Ray $60 = 3/4 of $80.

4
page 26

Answer: (a) blue (b) 9/25
Why: (a) Only yellow beads were added, so the 18 blue beads did not change. (b) Now there are 18 + 32 = 50 beads. 18/50 = 9/25.

Unit 4 · Triangle areas

Your turn
page 27

Answer: 54 cm²
Why: The right angle is at Q, so the sides PQ and QR meet at a right angle. Take QR as the base and PQ as the height. 1/2 × 12 × 9 = 54. The side PR = 15 cm is slanting, so it is not used.

START HERE

1
page 28

Answer: (a) 12 (b) 60
Why: Use the dashed height with the base it meets at a right angle. (a) 1/2 × 6 × 4 = 12. (b) 1/2 × 10 × 12 = 60. The slanting sides (5 and 5, 13 and 13) are not used.

2
page 28

Answer: (a) 24 (b) 6 (c) QR
Why: (a) 1/2 × 8 × 6 = 24. (b) PQ meets QR at the right angle Q, so the height is PQ = 6 cm. (c) With PQ as the base, the height is QR, because QR meets PQ at a right angle. PR is slanting.

3
page 28

Answer: (a) 8 (b) 10
Why: Base × height is 2 times the area. (a) 2 × 36 = 72 and 72 ÷ 9 = 8. (b) 2 × 45 = 90 and 90 ÷ 9 = 10.

4
page 28

Answer: (a) 42 (b) is the same
Why: (a) The base AB is 12 cm and the height is 7 cm, the distance between the 2 lines. 1/2 × 12 × 7 = 42. (b) Wherever P is on DC, the base is still 12 cm and the height is still 7 cm, so the area stays the same.

5
page 29

Answer: 84 cm²
Why: The square is 8 × 8 = 64 cm². The triangle is 1/2 × 8 × 5 = 20 cm². Add: 64 + 20 = 84 cm².

6
page 29

Answer: (a) 63 (b) 63
Why: (a) 1/2 × 14 × 9 = 63. (b) The rectangle is 14 × 9 = 126 cm². 126 - 63 = 63. The 2 white triangles also add to half of the rectangle.

7
page 29

Answer: 15 cm²
Why: The base AB is 6 squares long. The top point C is 5 squares above the base, so the height is 5. 1/2 × 6 × 5 = 15 cm².

8
page 29

Answer: 66 cm²
Why: Use BC as the base. The height from A meets the line BC extended at a right angle at H, and it is 12 cm. The base is only BC = 11 cm, not BH. 1/2 × 11 × 12 = 66. The slanting sides 20 and 13 are not used.

9
page 30

Answer: (a) 100 (b) 8
Why: (a) 10 × 10 = 100. (b) The triangle has area 100 too, so base × height = 200. 200 ÷ 25 = 8. Check: 1/2 × 25 × 8 = 100.

10
page 30

Answer: 98 cm²
Why: Think of the full rectangle 12 by 9 = 108 cm². The cut-off corner is a right-angled triangle. Its legs are 12 - 8 = 4 cm along the top and 9 - 4 = 5 cm down the right side. Its area is 1/2 × 4 × 5 = 10. The shape is 108 - 10 = 98 cm².

11
page 30

Answer: (a) 6 (b) 6
Why: Base × height = 18. The pairs are 1 and 18, 2 and 9, 3 and 6, 6 and 3, 9 and 2, 18 and 1: 6 pairs. A base of 3 goes with a height of 6.

12
page 30

Answer: (a) True (b) False (c) True
Why: (a) True: that is what makes it the height. (b) False: a slanting side does not meet the base at a right angle. (c) True: both have area 1/2 × 10 × 6 = 30 cm².

THINK HARDER

1
page 31

Answer: 4.8 cm
Why: The area does not depend on which side is the base. Use the 2 sides at the right angle: 1/2 × 8 × 6 = 24 cm². Now the base is 10 cm: 1/2 × 10 × height = 24, so 5 × height = 24 and the height is 4.8 cm.

2
page 31

Answer: 52.5 cm²
Why: The rectangle is 14 × 10 = 140 cm². White triangle ABM: 1/2 × 14 × 5 = 35. White triangle MCN: legs 5 and 7, 1/2 × 5 × 7 = 17.5. White triangle ADN: 1/2 × 10 × 7 = 35. White total: 35 + 17.5 + 35 = 87.5. Shaded: 140 - 87.5 = 52.5 cm².

3
page 31

Answer: (a) 24 (b) 12
Why: (a) Triangles ABD and ABC have the same height from A. The base BD is 1/3 of BC, so ABD is 1/3 of 72 = 24 cm². (b) Triangles BDE and ABD have the same height from B to the line AD. The base DE is half of DA, so BDE is half of 24 = 12 cm².

GO FURTHER

1
page 32

Answer: 112 cm²
Why: The square is 16 × 16 = 256 cm². White triangle ADE: 1/2 × 16 × 4 = 32. White triangle EBF: EB = 12 and BF = 8, so 1/2 × 12 × 8 = 48. White triangle FCD: 1/2 × 16 × 8 = 64. White total: 32 + 48 + 64 = 144. Shaded: 256 - 144 = 112 cm².

2
page 32

Answer: 13 cm²
Why: The rectangle round the triangle goes from 1 to 7 across and 1 to 6 up: 6 × 5 = 30 cm². Three right-angled triangles are outside the shape: 1/2 × 6 × 2 = 6, 1/2 × 4 × 3 = 6 and 1/2 × 2 × 5 = 5. Together 17. The triangle is 30 - 17 = 13 cm².

3
page 32

Answer: 8 cm²
Why: Join D to C. Triangles ADC and ABC have the same height from C, and AD is 1/4 of AB, so ADC = 1/4 of 96 = 24 cm². Now triangles ADE and ADC have the same height from D, and AE is 1/3 of AC, so ADE = 1/3 of 24 = 8 cm². The small triangle is 1/4 × 1/3 = 1/12 of the whole.

CHECK YOURSELF

1
page 33

Answer: 60 cm²
Why: The base is 24 cm and the height that meets it at a right angle is 5 cm. 1/2 × 24 × 5 = 60. The slanting sides (13 and 13) are not used.

2
page 33

Answer: 9 cm
Why: 2 × 54 = 108 is base × height. 108 ÷ 12 = 9. Check: 1/2 × 12 × 9 = 54.

3
page 33

Answer: 72 cm²
Why: The rectangle is 10 × 6 = 60 cm². The triangle on the right has a base of 6 cm (the side BC) and a height of 4 cm. 1/2 × 6 × 4 = 12. Total: 60 + 12 = 72 cm².

4 page 33
Answer: 15 bags
Why: The area is 1/2 × 18 × 10 = 90 m². 90 ÷ 6 = 15 bags.

Unit 5 · Volume vault

Your turn page 34
Answer: 13 cm
Why: The base area is 25 × 20 = 500 cm². The stone raises the water by 1500 ÷ 500 = 3 cm. The new depth is 10 + 3 = 13 cm.

START HERE

1 page 35
Answer: (a) 12 (b) 2 (c) 24, 24
Why: (a) 4 × 3 = 12 cubes in one layer. (b) The box is 2 cubes high, so 2 layers. (c) 12 × 2 = 24 cubes. Each cube is 1 cm³, so the volume is 24 cm³.

2 page 35
Answer: (a) 270 (b) 343
Why: (a) 9 × 6 × 5 = 54 × 5 = 270 cm³. (b) 7 × 7 × 7 = 49 × 7 = 343 cm³.

3 page 35
Answer: (a) 5 (b) 6
Why: (a) The base area is 8 × 5 = 40 cm². 200 ÷ 40 = 5 cm. (b) The base area is 144 ÷ 4 = 36 cm². 6 × 6 = 36, so each side is 6 cm.

4 page 35
Answer: (a) 3500 (b) 12 (c) 60
Why: (a) 3.5 × 1000 = 3500. (b) 12 000 ÷ 1000 = 12. (c) 50 × 40 × 30 = 60 000 cm³, and 60 000 ÷ 1000 = 60 litres.

5 page 36
Answer: (a) 14400 (b) 14.4 (c) 15.6
Why: (a) 40 × 30 × 12 = 1200 × 12 = 14 400 cm³. (b) 14 400 ÷ 1000 = 14.4 litres. (c) The full tank is 40 × 30 × 25 = 30 000 cm³ = 30 litres. 30 - 14.4 = 15.6 litres.

6 page 36
Answer: (a) 4 (b) 10
Why: The base area is 50 × 20 = 1000 cm². 4 litres = 4000 cm³. The rise is 4000 ÷ 1000 = 4 cm. The new depth is 6 + 4 = 10 cm.

7 page 36
Answer: 1080 cm³
Why: The rise is 11 - 8 = 3 cm. The volume of the stone is 24 × 15 × 3 = 360 × 3 = 1080 cm³.

8 page 36
Answer: (a) 5 (b) 15
Why: The base area is 60 × 30 = 1800 cm². 9 litres = 9000 cm³. The fall is 9000 ÷ 1800 = 5 cm. The new depth is 20 - 5 = 15 cm.

9 page 36
Answer: (a) 5 (b) 10
Why: List the sets of 3 edges with product 30: 1, 1, 30; 1, 2, 15; 1, 3, 10; 1, 5, 6; 2, 3, 5. That is 5 cuboids. Their totals are 32, 18, 14, 12 and 10. The smallest is 2 + 3 + 5 = 10 cm.

10 page 37

Answer: (a) 8 (b) 12 (c) 6 (d) 1
Why: (a) The 8 corner cubes have 3 painted faces. (b) Each of the 12 edges has 1 middle cube with 2 painted faces. (c) Each of the 6 faces has 1 middle cube with 1 painted face. (d) 1 cube is in the middle with no paint. Check: 8 + 12 + 6 + 1 = 27.

11 page 37

Answer: 30 cubes
Why: Along the length: 32 ÷ 6 = 5 and 2 cm is left over, so 5 cubes. Along the width: 20 ÷ 6 = 3, so 3 cubes. In height: 14 ÷ 6 = 2, so 2 layers. 5 × 3 × 2 = 30 cubes. (Not 8960 ÷ 216 = 41, because the cubes cannot be cut.)

12 page 37

Answer: 10 cm
Why: The water has a volume of 30 × 30 × 30 = 27 000 cm³. The base of the tank is 60 × 45 = 2700 cm². The depth is 27 000 ÷ 2700 = 10 cm.

13 page 37

Answer: 60 litres
Why: 7/10 - 2/5 = 7/10 - 4/10 = 3/10 of the tank is 18 litres. So 1/10 is 6 litres and the full tank is 10 × 6 = 60 litres.

14 page 37

Answer: (a) True (b) False (c) False
Why: (a) True. (b) The cuboid is 24 cm³. The cube is 27 cm³, so it is false. (c) False: 2 × 2 × 2 = 8, so the volume is 8 times as much.

THINK HARDER

1 page 38

Answer: 125 small cubes
Why: The cube has 6 faces. 54 ÷ 6 = 9 one-face cubes on each face. That is a 3 by 3 square in the middle of the face. So each face is 3 + 2 = 5 cubes across. The big cube is 5 × 5 × 5 = 125 small cubes.

2 page 38

Answer: 12 cm
Why: The water in A is 40 × 30 × 18 = 21 600 cm³. After pouring, this water is in both tanks at the same depth. The 2 bases are 1200 cm² and 600 cm², which is 1800 cm² together. The depth is 21 600 ÷ 1800 = 12 cm.

3 page 38

Answer: 12 cm
Why: The water volume is 40 × 20 × 9 = 7200 cm³. The block covers 10 × 20 = 200 cm² of the floor. The floor left for water is 800 - 200 = 600 cm². The depth is 7200 ÷ 600 = 12 cm. The block is 15 cm tall, so it stands above the water, and the answer is right.

GO FURTHER

1 page 39

Answer: 160 cm³
Why: The cube is 6 × 6 × 6 = 216 small cubes. 1 tunnel has 2 × 2 × 6 = 24 cubes. 3 tunnels have 72 cubes, but the 8 cubes in the middle (2 × 2 × 2) were counted 3 times. The tunnels take away 72 - 2 × 8 = 56 cubes. The solid has 216 - 56 = 160 cm³.

2 page 39

Answer: 15 minutes
Why: The tank holds 60 × 50 × 40 = 120 000 cm³ = 120 litres. Each minute the tank gains 9 - 1 = 8 litres. 120 ÷ 8 = 15 minutes.

3 page 39
Answer: 216 cm³
Why: Try edges 4, 6 and 9: 4 × 6 = 24, 4 × 9 = 36 and 6 × 9 = 54. They fit. The volume is 4 × 6 × 9 = 216 cm³. (Another way: the 3 areas multiplied are 24 × 36 × 54 = 46 656 = 216 × 216.)

CHECK YOURSELF

1 page 40
Answer: 240 cm³
Why: 12 × 5 × 4 = 60 × 4 = 240 cm³.

2 page 40
Answer: 12 cm
Why: 18 litres = 18 000 cm³. The base is 50 × 30 = 1500 cm². The depth is 18 000 ÷ 1500 = 12 cm.

3 page 40
Answer: 24 cubes
Why: The cube has 12 edges. On each edge the 2 middle cubes have 2 painted faces (the 2 end cubes are corners). 12 × 2 = 24.

4 page 40
Answer: 1200 cm³
Why: The rise is 14 - 10 = 4 cm. The volume is 20 × 15 × 4 = 300 × 4 = 1200 cm³.

Unit 6 · Rate racers

Your turn page 41
Answer: 66 bottles
Why: 24 ÷ 4 = 6 bottles in 1 minute. In 11 minutes: 6 × 11 = 66 bottles.

START HERE

1 page 42
Answer: 6, 30, 48
Why: In 3 minutes: 18 litres. In 1 minute: 18 ÷ 3 = 6 litres. In 5 minutes: 5 × 6 = 30 litres. In 8 minutes: 8 × 6 = 48 litres.

2 page 42
Answer: (a) 0.70 (b) 6.30
Why: (a) $4.20 ÷ 6 = $0.70. (b) 9 × $0.70 = $6.30.

3 page 42
Answer: (a) 9 (b) 225 (c) 30
Why: (a) 36 ÷ 4 = 9 rolls a minute. (b) 25 × 9 = 225. (c) 270 ÷ 9 = 30 minutes.

4 page 42
Answer: (a) 0.50 (b) 0.45 (c) Pack B
Why: (a) $2.50 ÷ 5 = $0.50. (b) $3.60 ÷ 8 = $0.45. (c) $0.45 is less than $0.50, so Pack B is the better buy.

5 page 43
Answer: (a) 12.00 (b) 11.25 (c) Shop Y
Why: 15 cans is 5 packs of 3 at X: 5 × $2.40 = $12.00. 15 cans is 3 packs of 5 at Y: 3 × $3.75 = $11.25. Shop Y is cheaper.

6 page 43
Answer: (a) 20 (b) 10
Why: (a) 14 + 6 = 20 litres a minute. (b) 200 ÷ 20 = 10 minutes.

7 page 43
Answer: 20 minutes
Why: The tank gains 25 - 10 = 15 litres each minute. 300 ÷ 15 = 20 minutes.

8
page 43

Answer: 15 minutes
Why: Together they print 40 + 20 = 60 pages a minute. 900 ÷ 60 = 15 minutes.

9
page 43

Answer: $4.40
Why: The first hour is $2.00. After that 1 hour 20 minutes is left. That is 2 full half hours and 20 minutes more, which counts as one more half hour: 3 lots. 3 × $0.80 = $2.40. Total: $2.00 + $2.40 = $4.40.

10
page 44

Answer: (a) 105 (b) 16
Why: (a) 9 × 15 = 135 litres are taken out. 240 - 135 = 105 litres are left. (b) 240 ÷ 15 = 16 minutes.

11
page 44

Answer: (a) 24 (b) 1440 (c) 1.44
Why: (a) 1 minute is 60 seconds, or 6 lots of 10 seconds. 6 × 4 = 24 ml. (b) 60 minutes: 60 × 24 = 1440 ml. (c) 1440 ÷ 1000 = 1.44 litres.

12
page 44

Answer: (a) True (b) False (c) True
Why: (a) True: double the cans, double the cost. (b) False: the larger pack can cost more for each item. (c) True: 5 + 7 = 12.

13
page 44

Answer: 5 visits
Why: Each visit costs Plan B $4 more than Plan A. The $20 monthly fee of Plan A is paid back after 20 ÷ 4 = 5 visits. At 5 visits: Plan A costs 20 + 20 = $40 and Plan B costs 5 × 8 = $40.

14
page 44

Answer: 22.5 litres
Why: 5 minutes is 300 seconds, which is 15 lots of 20 seconds. 15 × 1.5 = 22.5 litres. (Or: 3 × 1.5 = 4.5 litres in 1 minute, and 5 × 4.5 = 22.5.)

THINK HARDER

1
page 45

Answer: $18.50 (1 pack B and 2 packs C)
Why: Find the ways to make exactly 32 pens: 1 pack B + 2 packs C (8 + 24), 4 packs B (32), and 4 packs A + 1 pack C (20 + 12). Costs: $4.70 + 2 × $6.90 = $18.50; 4 × $4.70 = $18.80; 4 × $3.00 + $6.90 = $18.90. The least is $18.50.

2
page 45

Answer: 7.5 minutes (7 1/2)
Why: Pretend the job has 60 pages (a number that 12 and 20 both go into). P prints 60 ÷ 12 = 5 pages a minute. Q prints 60 ÷ 20 = 3 pages a minute. Together: 8 pages a minute. 60 ÷ 8 = 7.5 minutes.

3
page 45

Answer: 600 sheets
Why: 4 printers print 480 sheets in 6 minutes, so 1 printer in 6 minutes prints 480 ÷ 4 = 120 sheets. In 1 minute 1 printer prints 120 ÷ 6 = 20 sheets. 6 printers in 5 minutes: 6 × 5 × 20 = 600 sheets.

GO FURTHER

1
page 46

Answer: 8 litres a minute
Why: First half: 60 ÷ 6 = 10 minutes. Second half: 60 ÷ 12 = 5 minutes. The whole tank takes 15 minutes. The average rate is 120 ÷ 15 = 8 litres a minute. It is not (6 + 12) ÷ 2 = 9, because more time is spent at the slow rate.

2
page 46

Answer: 10 minutes (40 litres each)
Why: At first A has 80 litres more than B. Each minute A loses 5 and B gains 3, so the gap closes by 8 litres. 80 ÷ 8 = 10 minutes. Then A has 90 - 50 = 40 litres and B has 10 + 30 = 40 litres.

3
page 46

Answer: 13 minutes
Why: In the first 5 minutes tap A puts in 5 × 8 = 40 litres. 200 - 40 = 160 litres are left. Together the taps put in 8 + 12 = 20 litres a minute. 160 ÷ 20 = 8 more minutes. The tank is full 5 + 8 = 13 minutes after the start.

CHECK YOURSELF

1
page 47

Answer: 144 copies
Why: 54 ÷ 3 = 18 copies in 1 minute. 8 × 18 = 144 copies.

2
page 47

Answer: (a) 8 mangoes (b) 0.75
Why: 5 mangoes: $4.00 ÷ 5 = $0.80 each. 8 mangoes: $6.00 ÷ 8 = $0.75 each. $0.75 is less, so 8 mangoes is the better buy.

3
page 47

Answer: 10 minutes
Why: The tank gains 9 + 11 - 4 = 16 litres each minute. 160 ÷ 16 = 10 minutes.

4
page 47

Answer: $3.90
Why: The first hour is $1.50. 1 hour 45 minutes is left: 3 full half hours and 15 minutes more, so 4 lots. 4 × $0.60 = $2.40. Total: $1.50 + $2.40 = $3.90.

Unit 7 · Percentage puzzles

Your turn
page 48

Answer: $65.40
Why: 10% of $75 = $7.50, so 20% = $15 off. The cost after the discount is $60. 1% of $60 = $0.60, so 9% = $5.40. Mei pays $60 + $5.40 = $65.40.

START HERE

1
page 49

Answer: (a) 38 (b) 62
Why: (a) 38 of the 100 squares are shaded, so 38%. (b) 100 - 38 = 62 squares are not shaded, so 62%.

2
page 49

Answer: (a) 34 (b) 17 (c) 51
Why: 10% is $340 ÷ 10 = $34. 5% is half of that: $17. 15% = 10% + 5% = $34 + $17 = $51.

3
page 49

Answer: (a) 72 (b) 35
Why: (a) 18 ÷ 25 = 72 ÷ 100, so 72%. (b) 7 out of 20 = 35 out of 100 = 35%.

4 page 49
Answer: (a) 12 (b) 18 (c) 102
Why: (a) $120 ÷ 10 = $12. (b) 5% = $6, so 15% = $12 + $6 = $18. (c) $120 - $18 = $102.

5 page 50
Answer: (a) 5.76 (b) 69.76
Why: (a) 1% of $64 = $0.64. 9% = 9 × $0.64 = $5.76. (b) $64 + $5.76 = $69.76.

6 page 50
Answer: $72
Why: 20% of $150 = $30, so $120 is left. 40% of $120 = $48 for the book. $120 - $48 = $72. (Not 40% of $150!)

7 page 50
Answer: (a) 14 (b) 10 (c) 16
Why: (a) 10% of 40 = 4, 5% = 2, so 35% = 12 + 2 = 14. (b) 25% = 10 (a quarter of 40). (c) 40 - 14 - 10 = 16. (Check: 100% - 35% - 25% = 40%, and 40% of 40 = 16.)

8 page 50
Answer: (a) 70 (b) 68 (c) Quiz 1
Why: (a) 21 ÷ 30 = 7 ÷ 10 = 70%. (b) 34 out of 50 = 68 out of 100 = 68%. (c) 70% is more than 68%, so Quiz 1.

9 page 50
Answer: (a) 40 (b) 42 (c) Shop P
Why: (a) She pays for 2 T-shirts: 2 × $20 = $40. (b) 30% of $20 = $6 off each, so $14 each. 3 × $14 = $42. (c) $40 is less than $42, so Shop P.

10 page 51
Answer: (a) 60 (b) 48 (c) No
Why: (a) 20% of $50 = $10, so $60. (b) 20% of $60 = $12, so $60 - $12 = $48. (c) No. The second 20% is taken from $60, not from $50, so the cost does not come back to $50.

11 page 51
Answer: (a) 40 (b) 75
Why: (a) 2 out of 5 equal parts: 2/5 = 40/100 = 40%. (b) 3 out of 4 equal parts: 3/4 = 75%.

12 page 51
Answer: (a) True (b) True (c) False
Why: (a) 50% is a half: 40. (b) 90 ÷ 10 = 9. (c) False. Test with $100: 20% off gives $80. Then 10% of $80 is $8 off, so $72. A 30% discount gives $70.

13 page 51
Answer: $90
Why: 40% of $200 = $80 for shoes, so $120 is left. 25% of $120 = $30 for the gift. $120 - $30 = $90.

14 page 51
Answer: $292.50
Why: 1% of $250 = $2.50, so 9% = $22.50. The sofa costs $250 + $22.50 = $272.50. Add the delivery: $272.50 + $20 = $292.50.

THINK HARDER

1 page 52
Answer: (a) 144 (b) 28
Why: (a) 10% of $200 = $20, so the new cost is $180. 20% of $180 = $36, so the coat costs $180 - $36 = $144. (b) The total discount is $200 - $144 = $56. $56 out of $200 = 28 out of 100 = 28%. It is not 30%.

2 page 52
Answer: $75
Why: The bag now costs 80% of the old cost. 80% is $60, so 10% is $60 ÷ 8 = $7.50. The old cost is 100% = 10 × $7.50 = $75. Check: 20% of $75 = $15 and $75 - $15 = $60. (Adding 20% of $60 would give $72, which is wrong.)

3 page 52
Answer: $18
Why: The cost before GST is 100%. With GST it is 109%. 109% is $218, so 1% is $218 ÷ 109 = $2. The GST is 9 × $2 = $18. (9% of $218 would be $19.62, which is wrong because GST is 9% of the cost BEFORE GST.)

GO FURTHER

1 page 53
Answer: (a) 150 (b) 60
Why: (a) The change is $30. Out of the starting amount $20: 30/20 = 150/100 = 150%. (b) The change is $30 again. Out of the starting amount $50: 30/50 = 60/100 = 60%. The same $30 change gives different percentages because the starting amounts are different.

2 page 53
Answer: 40%
Why: Boys: 40% of 40 = 16. Girls: 24. Glasses: 25% of 16 = 4 boys and 50% of 24 = 12 girls. That is 16 children out of 40. 16/40 = 40/100 = 40%.

3 page 53
Answer: 50 blue beads
Why: Red beads: 40% of 50 = 20, and they do not change. After adding blue beads, red is 20% of all the beads. So 20 beads are 20%, and 100% is 100 beads. Blue now: 80. Blue at first: 30. Added: 80 - 30 = 50.

CHECK YOURSELF

1 page 54
Answer: (a) 18 (b) 9 (c) 27
Why: 10% = $180 ÷ 10 = $18. 5% = half of $18 = $9. 15% = $18 + $9 = $27.

2 page 54
Answer: $32.70
Why: 25% of $40 = $10, so the new cost is $30. 1% of $30 = $0.30, so 9% = $2.70. $30 + $2.70 = $32.70.

3 page 54
Answer: $30
Why: 25% of $80 = $20, so $60 is left. 50% of $60 = $30 for the gift. $60 - $30 = $30 is left.

4 page 54
Answer: Test 1
Why: 33 out of 50 = 66 out of 100 = 66%. 16 out of 25 = 64 out of 100 = 64%. 66% is higher, so Test 1.

Unit 8 · Angle chasers

Your turn page 55
Answer: 72°
Why: XY = XZ, so the angles at Y and Z are equal: angle Z = 54°. The angles of a triangle add up to 180°. Angle X = 180° - 54° - 54° = 72°.

START HERE

1
page 56

Answer: (a) 116 (b) 64 (c) 116
Why: (a) AOB is a straight line, so BOC = 180° - 64° = 116° (angles on a straight line). (b) BOD is opposite AOC, so BOD = 64° (vertically opposite angles are equal). (c) AOD is opposite BOC, so AOD = 116° (vertically opposite).

2
page 56

Answer: (a) 48 (b) 40
Why: (a) Angles on a straight line add up to 180°. 47° + 85° = 132°, and 180° - 132° = 48°. (b) Angles at a point add up to 360°. 125° + 90° + 105° = 320°, and 360° - 320° = 40°.

3
page 56

Answer: (a) 55 (b) 52
Why: The angles of a triangle add up to 180°. (a) 58° + 67° = 125°, and 180° - 125° = 55°. (b) The right angle is 90°. 90° + 38° = 128°, and 180° - 128° = 52°.

4
page 56

Answer: (a) 72 (b) 36
Why: (a) PQ = PR, so the angles opposite them are equal: R = Q = 72°. (b) The angles of a triangle add up to 180°. P = 180° - 72° - 72° = 36°.

5
page 57

Answer: (a) 110 (b) 70
Why: (a) Neighbouring angles of a parallelogram add up to 180°. B = 180° - 70° = 110°. (b) Opposite angles are equal. C = A = 70°.

6
page 57

Answer: (a) 28 (b) 56
Why: (a) A rhombus has 4 equal sides, so AB = BC and triangle ABC is isosceles. Angles BAC and BCA are equal. (180° - 124°) ÷ 2 = 28°. (b) Neighbouring angles of a rhombus add up to 180°. BAD = 180° - 124° = 56°.

7
page 57

Answer: (a) 70 (b) 105
Why: The angles between the parallel sides add up to 180°. (a) B and C are on the same slanting side: C = 180° - 110° = 70°. (b) A and D are on the other slanting side: D = 180° - 75° = 105°.

8
page 57

Answer: (a) 55 (b) 70 (c) 125 (d) 25
Why: (a) AB = AD, so ADB = ABD = 55° (equal base angles). (b) In triangle ABD: 180° - 55° - 55° = 70°. (c) BDC is a straight line: ADC = 180° - 55° = 125°. (d) In triangle ADC: 180° - 125° - 30° = 25°.

9
page 58

Answer: (a) 60 (b) 120
Why: (a) An equilateral triangle has 3 equal angles that add up to 180°: 180° ÷ 3 = 60°. (b) BCD is a straight line, so ACD = 180° - 60° = 120°.

10
page 58

Answer: (a) 40 (b) 55 (c) 2
Why: (a) The 2 equal angles are 70° and 70°. 180° - 140° = 40°. The angles are 70°, 70°, 40°. (b) The other two angles share 180° - 70° = 110°, so each is 55°. The angles are 70°, 55°, 55°. (c) Both triangles are possible, so there are 2.

11
page 58

Answer: (a) 120 (b) 120 (c) 150
Why: (a) 360° ÷ 60 = 6° each minute, and 20 × 6° = 120°. (b) The hour numbers are 30° apart (360° ÷ 12). At 4 o'clock the hands point at 12 and 4: 4 × 30° = 120°. (c) At 7 o'clock the hands are 7 × 30° = 210° apart one way. The smaller angle is 360° - 210° = 150°.

12
page 58

Answer: (a) angles of a triangle (b) angles on a straight line (c) neighbouring angles
Why: (a) P, Q and R are the 3 angles of a triangle: 55° + 75° + 50° = 180°. (b) The angles on a straight line add up to 180°. (c) A and B are neighbours, not opposite, and neighbouring angles of a parallelogram add up to 180°.

THINK HARDER

1
page 59

Answer: (a) 110 (b) 35 (c) 65
Why: (a) The angles between the parallel sides add up to 180°. D = 180° - 70° = 110°. (b) AD = DC, so triangle ADC is isosceles. DAC = DCA = (180° - 110°) ÷ 2 = 35°. (c) Angle C of the trapezium = 180° - 80° = 100°. ACB = 100° - 35° = 65°. Check in triangle ABC: CAB = 70° - 35° = 35°, and 35° + 80° + 65° = 180°.

2
page 59

Answer: (a) 76 (b) 44
Why: (a) Let each equal angle be 1 part. Then the top angle is 1 part + 24°. 3 parts + 24° = 180°, so 3 parts = 156° and 1 part = 52°. The top angle is 52° + 24° = 76°. (b) Let the top angle be 1 part. Each equal angle is 1 part + 24°. 3 parts + 48° = 180°, so 3 parts = 132° and 1 part = 44°. The top angle is 44°.

3
page 59

Answer: 20°
Why: A rhombus has 4 equal sides and ABE has 3 equal sides, so AD = AB = AE. Triangle ADE is isosceles. Angle DAE = DAB + BAE = 80° + 60° = 140°. The other 2 angles are equal: AED = (180° - 140°) ÷ 2 = 20°.

GO FURTHER

1
page 60

Answer: (a) 540 (b) 108
Why: (a) The 3 triangles fill the pentagon exactly, so the angles of the pentagon are the angles of the 3 triangles: 3 × 180° = 540°. (b) All 5 angles are equal: 540° ÷ 5 = 108°.

2
page 60

Answer: 10°
Why: From 12: the hour hand is at 4 × 30° = 120° at 4 o'clock, and 20 minutes later it has moved 20 × 0.5° = 10° more, so it is at 130°. The minute hand is at 20 × 6° = 120°. The angle between them is 130° - 120° = 10°.

3
page 60

Answer: 36°
Why: Let angle A = 1 unit. AD = DC, so angle DCA = 1 unit. Angle BDC = 2 units (the 2 angles on the other side add up to it: angles of a triangle, or a straight line ADB). DC = CB, so angle DBC = 2 units. AB = AC, so angle ACB = angle ABC = 2 units. Then angle DCB = 2 - 1 = 1 unit. Triangle DBC has angles 2, 2 and 1 units, which add up to 180°: 5 units = 180°, so 1 unit = 36°. Angle A = 36°.

CHECK YOURSELF

1
page 61
Answer: 47°
Why: AOB is a straight line, so COB = 180° - 133° = 47° (angles on a straight line).

2
page 61
Answer: 40°
Why: The angles of a triangle add up to 180°. Q and R are equal. 180° - 100° = 80°, and 80° ÷ 2 = 40°.

3
page 61
Answer: 115°
Why: Opposite angles of a parallelogram are equal. D is opposite B, so D = 115°. (Neighbouring angles add up to 180°: A = 65°, and D = 180° - 65° = 115°.)

4
page 61
Answer: 22°
Why: AB = AD, so ADB = ABD = 62°. BDC is a straight line, so ADC = 180° - 62° = 118°. In triangle ADC: ACD = 180° - 118° - 40° = 22°.

Unit 9 · Pattern generalisers

Your turn
page 62
Answer: 61 sticks
Why: The first hexagon needs 6 sticks. Each new hexagon shares a side, so it adds 5 sticks. Rule: sticks = 5 × hexagons + 1. For 12 hexagons: 5 × 12 + 1 = 61 sticks.

START HERE

1
page 63
Answer: (a) 21 (b) 26 (c) 5
Why: Figure 1 has 6 sticks and figure 2 has 11 sticks (figure 3 has 16). Each new hexagon adds 5 sticks because 1 side is shared. (a) 16 + 5 = 21. (b) 21 + 5 = 26. (c) 5.

2
page 63
Answer: (a) 5, 1 (b) 101
Why: (a) The first hexagon has 6 sticks, which is 5 + 1. Each new one adds 5. So sticks = 5 × hexagons + 1. (b) 5 × 20 + 1 = 101. (Not 2 × 51: doubling counts the starting stick twice.)

3
page 63
Answer: (a) 19 (b) 19, 4
Why: (a) 96 - 1 = 95, and 95 ÷ 5 = 19 hexagons. (b) 100 - 1 = 99, and 99 ÷ 5 = 19 with 4 left over. So 19 hexagons use 96 sticks and 4 sticks are left.

4
page 63
Answer: (a) 27, 32 (b) 52 (c) No
Why: Each number is 5 more than the one before. (a) 27 and 32. (b) There are 9 jumps of 5 from the 1st to the 10th: 7 + 9 × 5 = 52. (c) 100 - 7 = 93, and 93 is not a multiple of 5, so 100 is not in the pattern. (Every number ends in 2 or 7.)

5
page 64
Answer: (a) 17 (b) 61 (c) 4, 1 (d) 20
Why: Each figure has 4 arms of n tiles and 1 middle tile. (a) 4 × 4 + 1 = 17. (b) 4 × 15 + 1 = 61. (c) The rule is tiles = 4 × figure number + 1. (d) 81 - 1 = 80, and 80 ÷ 4 = 20.

6
page 64

Answer: (a) 8 (b) 7
Why: Try 10 adults and 5 children: $90 + $25 = $115, too much. Try 8 adults and 7 children: $72 + $35 = $107. Right. (Another way: all 15 child tickets cost $75. Each adult ticket instead of a child ticket adds $4. $107 - $75 = $32, and 32 ÷ 4 = 8 adult tickets.)

7
page 64

Answer: (a) 10 (b) 7
Why: Suppose all 17 are stools: 17 × 3 = 51 legs. That is 61 - 51 = 10 legs short. Each chair instead of a stool adds 1 leg, so there are 10 chairs. Then 17 - 10 = 7 stools. Check: 40 + 21 = 61.

8
page 64

Answer: (a) 3 (b) 9
Why: All 12 right would score 12 × 5 = 60 marks. Each wrong answer costs 5 marks and also loses 2 marks: 7 marks. 60 - 39 = 21, and 21 ÷ 7 = 3 wrong. Then 12 - 3 = 9 right. Check: 45 - 6 = 39.

9
page 65

Answer: (a) 22 (b) 4, 2 (c) 12
Why: 1 table has 6 chairs and 2 tables have 10. Each new table adds 4 chairs (2 on each side). (a) 4 × 5 + 2 = 22. (b) chairs = 4 × tables + 2. (c) 50 - 2 = 48, and 48 ÷ 4 = 12 tables.

10
page 65

Answer: (a) 3, 4 (b) 40 (c) 32
Why: Each figure adds 3 sticks. Figure 1 has 7 = 3 + 4. (a) sticks = 3 × figure number + 4. (b) 3 × 12 + 4 = 40. (c) 100 - 4 = 96, and 96 ÷ 3 = 32.

11
page 65

Answer: (a) True (b) False (c) True
Why: (a) 101 - 5 = 96 and 96 ÷ 4 = 24, so it is the 25th number: true. (b) 20 hexagons need 5 × 20 + 1 = 101 sticks, not 102: false. (c) The numbers end in 6, 1, 6, 1, ... so none ends in 3: true.

THINK HARDER

1
page 66

Answer: 16 coins
Why: Suppose all 24 coins are 20¢ coins: 24 × 20¢ = $4.80. That is $9.60 - $4.80 = $4.80 short. Each 50¢ coin instead of a 20¢ coin adds 30¢. $4.80 ÷ 30¢ = 16 coins. So 16 fifty-cent coins and 8 twenty-cent coins. Check: $8.00 + $1.60 = $9.60.

2
page 66

Answer: (a) 55 (b) 14
Why: Figure 10 is 1 + 2 + 3 + ... + 10. Two copies of the staircase fit together as a rectangle 10 by 11 = 110 tiles. So one staircase has 55. (b) For figure 14: 14 × 15 = 210 and 210 ÷ 2 = 105. Figure 13 has 91 tiles and figure 15 has 120, so only figure 14 has 105.

3
page 66

Answer: 625
Why: 1 = 1 × 1, 1 + 3 = 4 = 2 × 2, 1 + 3 + 5 = 9 = 3 × 3. Each new odd number adds an L-shaped layer to make the next square. The odd numbers up to 49 are 25 numbers (49 = 2 × 25 - 1). So the total is 25 × 25 = 625.

GO FURTHER

1
page 67

Answer: 1716
Why: From 4 to 100 there are (100 - 4) ÷ 3 = 32 jumps, so 33 numbers. Pair them: 4 + 100 = 104, 7 + 97 = 104, and so on. Each pair makes 104. There are 33 numbers, which is 16 pairs and a middle number 52. 16 × 104 + 52 = 1716. (Or: 33 × 104 ÷ 2 = 1716.)

2
page 67

Answer: (a) 10 (b) 66
Why: (a) Each of the 5 people shakes 4 hands: 5 × 4 = 20. Each handshake is counted twice (once for each person), so 20 ÷ 2 = 10. (b) 12 × 11 = 132, and 132 ÷ 2 = 66 handshakes.

3
page 67

Answer: 20 right
Why: Ali answers 30 - 3 = 27 questions. All 27 right would score 108 marks. Each wrong answer loses 4 marks and has 1 mark taken away: 5 marks. 108 - 73 = 35, and 35 ÷ 5 = 7 wrong. Right: 27 - 7 = 20. Check: 80 - 7 = 73.

CHECK YOURSELF

1
page 68

Answer: 76
Why: Each number is 5 more. There are 14 jumps of 5 from the 1st to the 15th: 6 + 14 × 5 = 76.

2
page 68

Answer: Figure 13
Why: 41 - 2 = 39, and 39 ÷ 3 = 13. Check: 3 × 13 + 2 = 41.

3
page 68

Answer: 10 chairs
Why: Suppose all 20 are stools: 60 legs. That is 70 - 60 = 10 legs short. Each chair adds 1 leg, so there are 10 chairs (and 10 stools).

4
page 68

Answer: 25 hexagons
Why: 130 - 1 = 129, and 129 ÷ 5 = 25 with 4 left over. So 25 hexagons use 126 sticks, and 4 sticks are left.

Unit 10 · Challenge day

Your turn
page 69

Answer: Kumar
Why: Ben cannot have juice or milk, so Ben has tea. Kumar does not choose milk and tea is taken, so Kumar has juice. Aisha has the milk. Check: Aisha does not have tea. All clues are true.

START HERE

1
page 70

Answer: 10
Why: Work backwards, undoing the last step first. Undo halve: 19 × 2 = 38. Undo add 8: 38 - 8 = 30. Undo multiply by 3: 30 ÷ 3 = 10. Check: 10 × 3 = 30, 30 + 8 = 38, 38 ÷ 2 = 19.

2
page 70

Answer: $28
Why: Work backwards. Before the $6 he had $15 + $6 = $21. The $21 is the 3/4 left after spending 1/4. So 1/4 is $7 and the whole is 4 × $7 = $28. Check: 1/4 of 28 = 7, 28 - 7 = 21, 21 - 6 = 15.

3
page 70

Answer: Aisha
Why: Ravi owns neither the parrot nor the goldfish, so Ravi owns the turtle. Aisha does not own the goldfish and the turtle is taken, so Aisha owns the parrot. Wei owns the goldfish.

4
page 70

Answer: (a) 33 (b) 19
Why: Take away the 14: 52 - 14 = 38. Share equally: 19 and 19. Add the 14 back to one of them: 33 and 19. Check: 33 + 19 = 52 and 33 - 19 = 14.

5
page 70

Answer: (a) 9 (b) 4 (c) 1 (d) 14
Why: (a) The grid has 9 small squares. (b) A 2 by 2 square can start in 2 columns and 2 rows: 2 × 2 = 4. (c) There is 1 big square. (d) 9 + 4 + 1 = 14.

6
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Answer: Ali
Why: Try each child. If Ali broke it: Ali lies, Ben tells the truth, Siti lies (Ben is not lying). That is 1 truth. Good. If Ben broke it: Ali is true, Ben lies, Siti is true. That is 2 truths, which is wrong. If Siti broke it: Ali is true, Ben lies, Siti is true: 2 truths, wrong. Only Ali fits.

7
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Answer: 8 years
Why: The difference in their ages is always 40 - 8 = 32 years. When Dad is 3 times as old as Ali, the difference is 2 times Ali's age, so Ali is 16. That is 16 - 8 = 8 years from now. Check: Dad 48 = 3 × 16.

8
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Answer: 36 stickers
Why: Work backwards. The 12 left are half of the rest, so the rest was 24. The 24 is 2/3 of his stickers, so 1/3 is 12 and he had 36. Check: 36 - 12 = 24, 24 - 12 = 12.

9
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Answer: (a) 8 (b) 12
Why: Suppose all 20 are hens: 40 legs. That is 56 - 40 = 16 legs short. Each cow instead of a hen adds 2 legs, so 16 ÷ 2 = 8 cows. Then 20 - 8 = 12 hens. Check: 32 + 24 = 56.

10
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Answer: (a) 8 and 9
Why: List the pairs with product 72: 1 and 72, 2 and 36, 3 and 24, 4 and 18, 6 and 12, 8 and 9. Their sums are 73, 38, 27, 22, 18 and 17. Only 8 and 9 have a sum of 17.

11
page 72

Answer: (a) 6 (b) 132 and 312 (c) 231
Why: List in order: 123, 132, 213, 231, 312, 321. That is 6 numbers. (b) Even numbers end in 2: 132 and 312. (c) The differences from 250 are 127, 118, 37, 19, 62 and 71. The smallest is 19, so 231.

12
page 72

Answer: 16 times
Why: In the ones place: 5, 15, 25, 35, 45 and 55 is 6 times. In the tens place: 50 to 59 is 10 times. 6 + 10 = 16. (55 has the digit 5 twice and is counted in both places.)

THINK HARDER

1
page 73

Answer: (a) swimming (b) badminton (c) tennis (d) football
Why: Wei does not play swimming, badminton or tennis, so Wei plays football (clue 3). Ben does not play tennis or badminton, and football is taken, so Ben plays swimming. Kumar does not play swimming or tennis, so Kumar plays badminton. Aisha plays tennis. Check all 3 clues: each is true.

2
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Answer: (a) 11 (b) 30
Why: (a) 1 by 1: 8. 2 by 2: 3 (3 places across, 1 place up). Total 8 + 3 = 11. (b) Count by size, width by height. Height 1: widths 1, 2, 3, 4 give 8, 6, 4, 2 rectangles = 20. Height 2: widths 1, 2, 3, 4 give 4, 3, 2, 1 = 10. Total 30. Check another way: choose 2 of 3 horizontal lines (3 ways) and 2 of 5 vertical lines (10 ways): 3 × 10 = 30.

3
page 73

Answer: 24 customers
Why: Work backwards. After the last step 14 are left, and that is 2/3 of the number before it, so there were 14 ÷ 2 × 3 = 21. Before 3 arrived: 21 - 3 = 18. The 18 is 3/4 of the first number, so 1/4 is 6 and at first there were 24. Check: 24 - 6 = 18, 18 + 3 = 21, 21 - 7 = 14.

GO FURTHER

1
page 74

Answer: 28 numbers
Why: Sort by the hundreds digit. If it is 1, the other 2 digits add up to 6: 06, 15, 24, 33, 42, 51, 60 is 7 numbers. If it is 2, the others add up to 5: 6 numbers. If it is 3, add up to 4: 5 numbers. If 4: 4 numbers. If 5: 3 numbers. If 6: 2 numbers. If 7: 1 number (700). Total: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28.

2
page 74

Answer: (a) 11 (b) 11 (c) 2
Why: A 10¢ coin and a 50¢ coin make a pair worth 60¢. Suppose all 24 coins are 20¢: $4.80. That is $2.20 short. Replace two 20¢ coins (40¢) with 1 pair (60¢): it adds 20¢. $2.20 ÷ 20¢ = 11 pairs. So 11 coins of 50¢, 11 coins of 10¢ and 24 - 22 = 2 coins of 20¢. Check: $5.50 + $1.10 + $0.40 = $7.00.

3
page 74

Answer: 4 socks
Why: In the worst case you take 1 red, 1 blue and 1 black sock: 3 socks and no pair. The 4th sock must match one of them. So 4 socks make sure of a pair. (3 socks are not enough because they could be 3 different colours.)

CHECK YOURSELF

1
page 75

Answer: 8
Why: Work backwards. Undo multiply by 4: 60 ÷ 4 = 15. Undo add 7: 15 - 7 = 8. Check: 8 + 7 = 15, 15 × 4 = 60.

2
page 75

Answer: $35
Why: Before the $9 she had $12 + $9 = $21. The $21 is 3/5 of her money, so 1/5 is $7 and the whole is 5 × $7 = $35. Check: 2/5 of 35 = 14, 35 - 14 = 21, 21 - 9 = 12.

3
page 75

Answer: 4 goats
Why: Suppose all 9 are hens: 18 legs. That is 26 - 18 = 8 legs short. Each goat adds 2 legs, so 8 ÷ 2 = 4 goats. Check: 4 × 4 + 5 × 2 = 16 + 10 = 26.

4
page 75

Answer: 8 squares
Why: There are 6 small squares. A 2 by 2 square can go in 2 places. 6 + 2 = 8.